

REPORT FRONT SHEET

MOLECULAR BIOLOGY, Master Course EXPERIMENTAL CELL BIOLOGY

NAME OF EXPERIMENT: Exercise 1 cell counting and NaBu treatment

DATE FOR PERFORMANCE OF EXPERIMENT: 15-06-2026 -- 26-06-2026

TEACHER: Ole Vang

TEAM Green

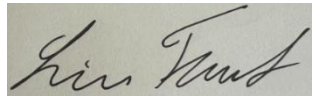
GROUP NUMBER: 3

Names and signatures

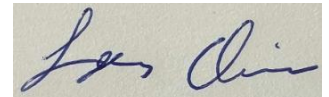
Andreas Giovanni Kristian
Frilev Olesen,



Line Babette Trust,



Lars Eivind Bustamante
Christoffersen



DATE(S) FOR HANDING IN THE REPORT TO THE TEACHER:

Date for 1. Time
26.06.2026

Date for 2. time

Date for 3. Time

TEACHER COMMENTS:

There are several corrections / changes that are needed to be made. Mostly in the result section

REPORT APPROVED BY TEACHER:

Signature

Date

This is the front page for the reports and has to be attached to the report. The original front page should be attached when a revised report is handed in.

THE BIOLOGY STUDY BOARD

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Introduction

Mammalian cell cultures are considered one of the major tools of cell biology, as they enable precise control over experimental conditions necessary to study proliferation, differentiation and chemical effect on key cellular processes. Because of this, cell cultures are a key in vitro way of investigating the effect of bioactive substances on cancer cells gene expression and physiology.

The experiment involves the use of two colorectal cancer cell lines from humans. DLD-1 being highly proliferative and undifferentiated and CaCO-2 coming from adenocarcinoma, known to differentiate spontaneously.

The purpose of this research is to maintain cell cultures, freezing and recovering cell cultures. DLD-1 and CaCO-2 have both been cultured and tested in 24-well plates. Coulter counter and hemocytometer were used to count cells. Phase-contrast microscopy was applied to observe changes in cell morphology, including cell size and other structural changes.

Two chemicals were tested for their biological effects with dose response studies. One was the fatty acid known to inhibit histone deacetylase and gene expression modulation, sodium butyrate or NaBu. This chemical should have an effect on regulating cell proliferation and differentiation. The other compound used is resveratrol, known to have many characteristics, however, for this experiment we focus on its anti-proliferative effect and apoptotic effects.

Preliminary experiments were done to ensure that concentrations of NaBu would have considerable effects without compromising the viability when using it through different stages and including mixtures with resveratrol.



Methods

Cell Culturing

Human colorectal adenocarcinoma cell lines Caco-2 and DLD-1 were cultured under sterile conditions during the experiment. The Caco-2 cell line was grown in Dulbecco's Modified Eagle Medium (DMEM). On the other hand, the DLD-1 cell line was cultured in McCoy's 5A medium. Both media were enriched with 10% Fetal Bovine Serum (FBS) and 1% Penicillin-Streptomycin. The cells were maintained at 37°C in a humidified environment with 5% CO₂.

They were subcultured before attaining full confluence in order to maintain exponential growth. Human colorectal adenocarcinoma cell lines Caco-2 and DLD-1 were cultured under sterile conditions during the experiment. The Caco-2 cell line was grown in DMEM. On the other hand, the DLD-1 cell line was cultured in McCoy's 5A medium. Both media were enriched with 10% FBS and 1% Penicillin-Streptomycin. The cells were maintained at 37°C in a humidified environment with 5% CO₂. They were subcultured before attaining



Cell harvesting

Adherent cells were detached after aspirating the medium and two washes. DLD-1 cultures were washed with PBS without calcium and magnesium while Caco-2 cultures were washed with sterile 85 mM sodium citrate solution. Cells were digested by adding 0.05% trypsin-EDTA and waited for microscopic visualization of cell detachment. The digestion process was stopped by adding fresh growth media, and the suspension was collected and mixed properly.

The cell concentration and distribution were measured by Beckman Coulter Multisizer 4e. Before counting, 200-500 µL of cell suspension was diluted with 10 mL isotonic saline (0.9% NaCl). Before measuring, the parameters of the instrument were set and then the data obtained from counting were exported for further analysis by Multisizer program.



Cell freezing

The DLD-1 cells that had reached the log phase of growth were trypsinized and spun down at $300 \times g$ for 3-5 minutes. This cell pellet was then resuspended in the freezing solution, which comprised of McCoy's 5A containing 10% fetal bovine serum, 1% penicillin-streptomycin, and 10% DMSO. Cell suspensions were dispensed into two cryovials with approximately 2 million cells each, and then slowly brought down to -80°C before being stored.

To thaw the cells, the vials were taken out of storage, transported to a lab bench in ice and then thawed in a water by repeated washing with pre-warmed culture medium.



Cell Dose Response

For the first week of experimentation, Caco-2 and DLD-1 cells were plated on 24-well plates in 1 mL of complete medium at a seeding density of 2.5×10^4 cells/well. The next day, fresh medium containing different concentrations of sodium butyrate (NaBu) 0, 0.1, 1, or 10 mM – was added to cells, while control wells had corresponding media only.



The cellular morphology was observed 48 hours after treatment using phase contrast microscopy. Pictures were taken in the center of each well without prolonging the plate stay in the incubator. After that, cells were trypsinized, suspended in isotonic saline solution, and counted in the Coulter Counter. Imaging was done on a phase-contrast microscope and were captured at 20x resolution. ImageJ was used to manually segment cells and for calculating their areas. In the image post processing step we forgot to set the pixel to μm scale using the leica image, however inspecting the images we saw that the default unit was 300 pixels/inch, while the leica scale gave 2.96 pixels/ μm . Thus, taking the areas and multiplying them by the factor $(300 \text{ pixels/inch})^2 / (296 \text{ pixels}/\mu\text{m})^2$ yielded the correct scale for the images when processing the data in R. Cells areas were estimated on groups of six cells for each image except Caco-2 well A1 and D3 where individual cells were used.

Cell Treatment with Resveratrol and Sodium Butyrate

The second week saw the performance of dose-response experiments on resveratrol, alone and in combination with sodium butyrate. Cells were seeded in 2x 24 well plates at the same density as in the week 1 experiment, allowing for overnight cell adhesion before treatment.

Treatment with resveratrol was carried out at concentrations of 2.5, 5, 10, 20, and 40 μM . Combination of resveratrol with sodium butyrate entailed the use of the selected concentration of the latter along with each of resveratrol concentrations. Equal concentrations of DMSO and PBS were applied to compensate for any solvent effects.

After 48 hours of treatment, phase-contrast microscopy was performed. Next, cells were collected, counted, and characterized in accordance with the counting procedure used during the experiment with sodium butyrate.

Cell Measurement

Cell morphology changes were measured by ImageJ. First, images obtained with a microscope were scaled against 200 μm reference line. Borders of each individual cell were manually marked to measure its area. Around ten cells for each treatment condition were analyzed when possible. Values of the diameters received from the Coulter Counter measurements were used in addition to imaging data to obtain results through two independent methods.

Results

Sub cultivation and growth rate

DLD-1	Cell-concentration	Notes
15-june-2026	885.870 cells/ml	
16-june-2026		20% confluence, attached to the bottom of the flask, looks healthy.
17-june-2026		35% confluence started to proliferate.
18-june-2026	497.913 cells/ml	50% confluence, with individual groups of around 10-30 cells. Splitting cells. Coulter counter was used to determine cell-number and 500.000 cells was seeded in new T75flask.
22.-june-2026	1.131.180 cells/ml	New passage with 500.000 cells pr flask
23-june-2026		30 % confluence. Started to proliferate.
24-june-2026		40% confluence. Media looks light orange
25-june-2026	208.998 cells/ml	70% confluence. Splitting cells. Coulter counter was used to determine cell-number and 500.000 cells was seeded in new T75flask. got all cells loose during trypsinization, but many cells were still present in the flask, and therefore not that many in the cell count on coulter counter.

Table 1. Summary of DLD-1 cell culture maintenance from 15 to 25 June 2026, including cell concentration measured using a Coulter counter (cells/mL), estimated cell confluence, observations of cell morphology and growth.

On-going cell culture for DLD-1 cell line

The DLD-1 cells attached well after splitting and seeding, throughout the period. There was active proliferation and the cells was easy to manage and predict. The main challenge occurred during splitting and trypsinization on June 25. Although the cells was detached from the flask, many cells remained in the flask, and this resulted in a lower coulter counter measurement (208.998 cells/ml).

Caco-2	Cell concentration	Notes
15-june-2026	1.212.372 cells/ml	Splitting cells. Coulter counter was used to determine cell-number and 500.000 cells was seeded in new T75flask.
16-june-2026		20% confluence
17-june-2026		30% confluence
18-june-2026	116.688 cells/ml	70% confluence. Splitting cells. Coulter counter was used to determine cell-number and 500.000 cells was seeded in new T75flask.
22.-june-2026	195.075 cells/ml (new flask)	Cells was to confluent. Marianne provided us with a new T75-flask
23-june-2026		
24-june-2026		45% confluence
25-june-2026		70-80% confluence Cells was splitted and counted on the coulter counter. 300.000 cells was seeded in new T75 flask.
26-june-2026		Cells was splitted 1:1 to prevent them from growing to much doing the weekend.

Table 2. Summary of Caco-2 cell culture maintenance from 15 to 26 June 2026, including cell concentration measured using a Coulter counter (cells/ mL), estimated cell confluence, observations of cell morphology and growth, and details of passaging.

On-going cell culture for Caco-2 cell line.

The Caco-2 cells attached successfully after splitting and seeding. They were a little more challenging to handle since they were “slow starters”, and then suddenly just proliferated fast. On 22 June our on-going culture became to confluent and we were therefore supplied with a new flask from Marianne.

Determination of Growth rate

We estimated the doubling time for DLD-1 to be 21,3 hours based on our calculations and the doubling time for Caco-2 to be 48.9 hours (see appendix).



Freezing-thawing procedure

DLD-1 cells were frozen on the 18th june 2026 and were thawed on the 24th june 2026. On the 25th June 2026, the cells were approximately 20% confluent as seen in figure 1, and one day later they had reached a much higher confluence showing that the cells were very viable after the freeze-thawing procedure.



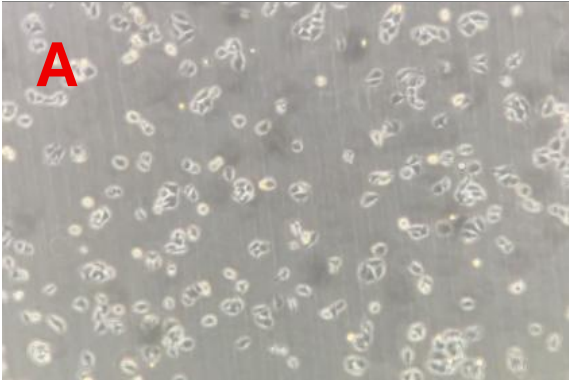
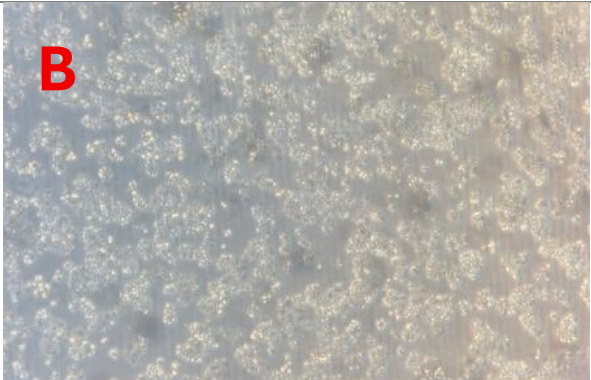
Thawed cells	
	
Picture taken of thaws cells on 25 June 2026. Taken through microscope at 10x objective.	Picture taken of thawed cells on 26 June 2026. Taken through microscope at 10x objective.

Figure 1. Representative microscopic images of thawed cells following recovery after thawing. (A) Cells on 25 June 2026 and (B) cells on 26 June 2026. Images were captured using a light microscope at 10× magnification to evaluate cell morphology and attachment.

Cell diameter from coulter counter

To determine the mean cell diameter coulter counter was used after 48 hours of treatment with increasing concentration of NaBu. As displayed in figure 2 a concentration-dependent increase in cell diameter was observed. The average cell diameter increased from 16,1 μm in the control sample (PBS) to 18.2 μm , 19.0 μm and 20.0 μm following treatment with 0.5, 1 and 5 mM NaBu, respectively.

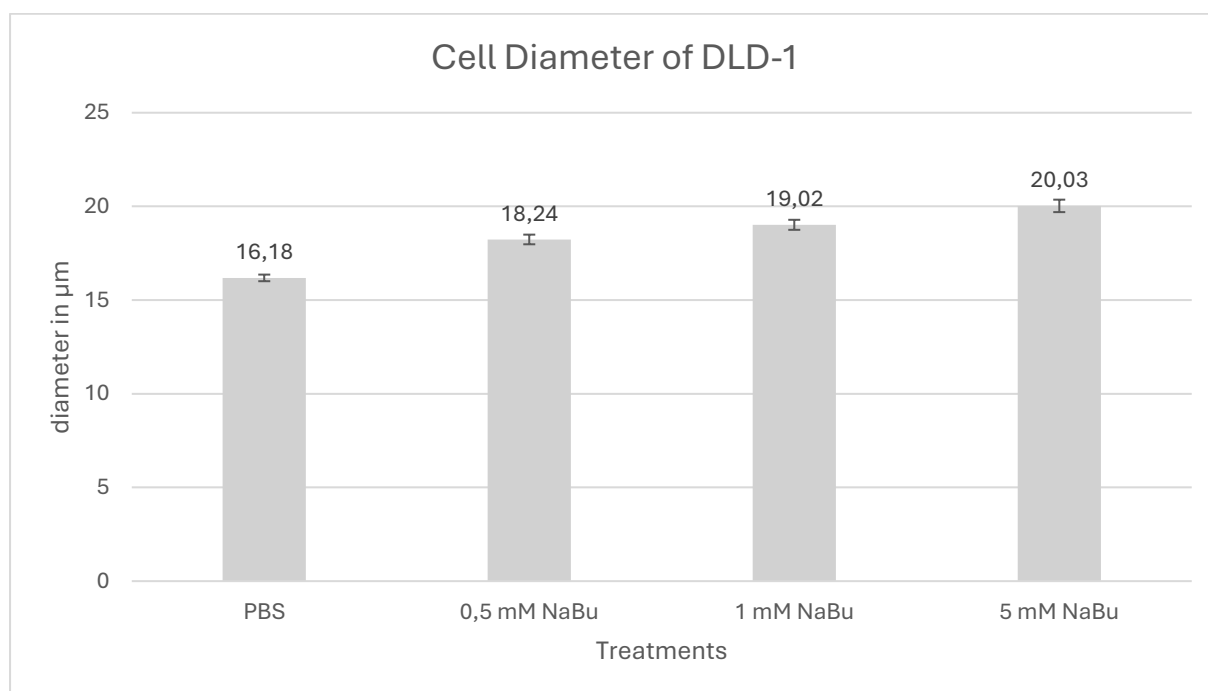


Figure 2. Effect of NaBu on DLD-1 cell diameter measured by Coulter counter. Mean cell diameter in μm of DLD-1 cells after 48 h treatment with PBS (control), 0.5, 1 and 5 mM sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of four technical replicates.

Caco-2 cells showed only minor changes in cell diameter after NaBu treatment (Figure 3). Mean diameters remained close to 20 μm for all treatment groups, with only a slight decrease observed at 1 mM NaBu. Furthermore the standard deviations from Caco-2 cells was a lot higher than observed for DLD-1 cells on figure x above.

Overall, NaBu had a considerably stronger effect on cell size in DLD-1 cells than in Caco-2 cells.

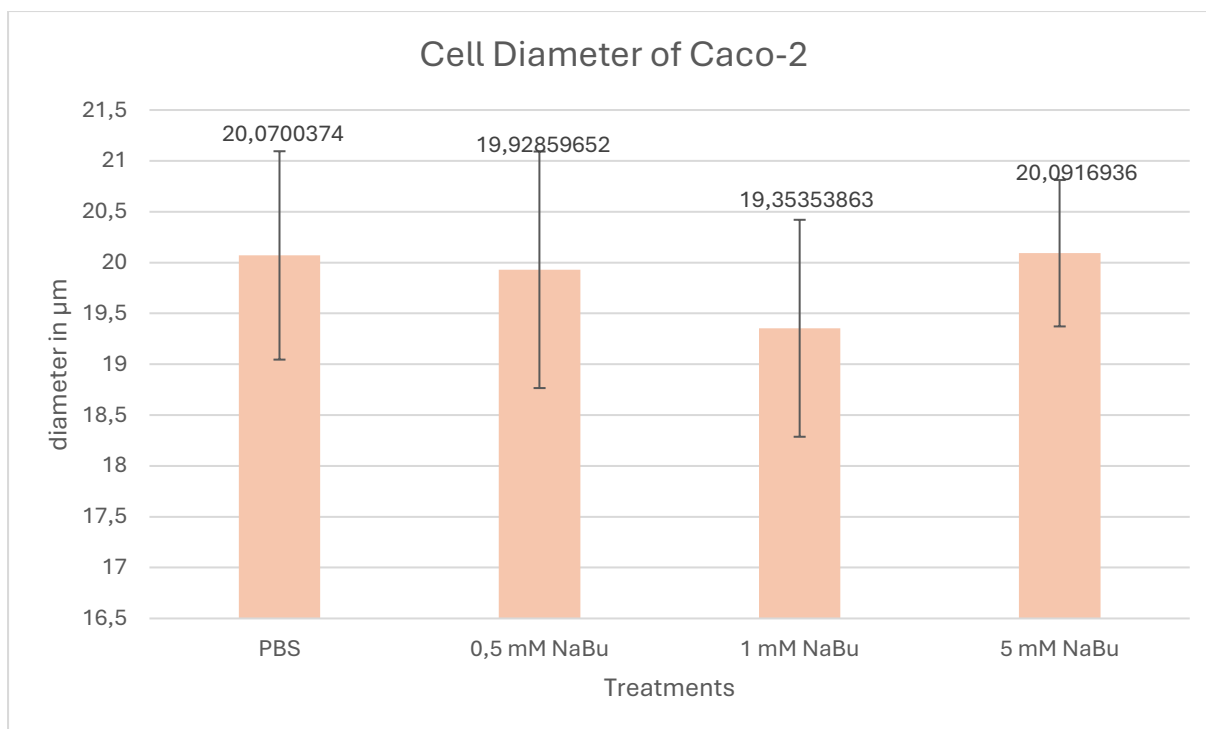


Figure 3. Effect of NaBu on Caco-2 cell diameter measured by Coulter counter.

Mean cell diameter in μm of Caco-2 cells after 48 h treatment with PBS (control), 0.5, 1 and 5 mM sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of three technical replicates.



The effect of resveratrol with and without NaBu on DLD-1 cell diameter is displayed on figure 4 below. Treatment with resveratrol alone, caused a change in cell diameter ranging from 16 μM to 21 μM to at the highest concentration of Resveratrol.

When NaBu was added in combination with Resveratrol. The cell diameter was generally higher with NaBu, except for the highest concentration of resveratrol at 40 μM .

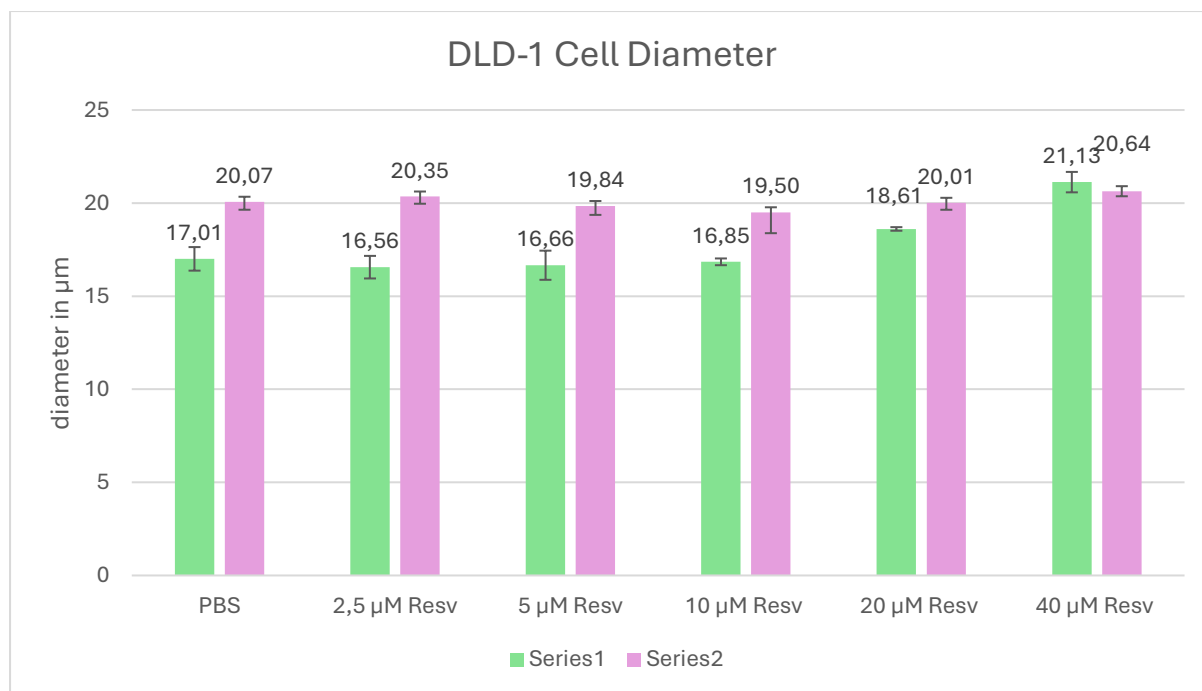


Figure 4. Effect of resveratrol with and without NaBu on DLD-1 cell diameter measured by Coulter counter. Mean cell diameter in μm of DLD-1 cells following treatment with increasing concentrations of resveratrol (2.5–40 μM) in the absence (green) or presence (purple) of sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of four technical replicates.

Cell count from coulter counter data

To see the effect of different NaBu concentrations on DLD-1 and CaCO₂, the data gathered from the first week coulter counter was inserted into a calculation template provided to us by our instructor. The template provided us with a mean and standard deviation that could be plotted into a graph to visualize the effect of NaBu to the cell count of DLD-1.

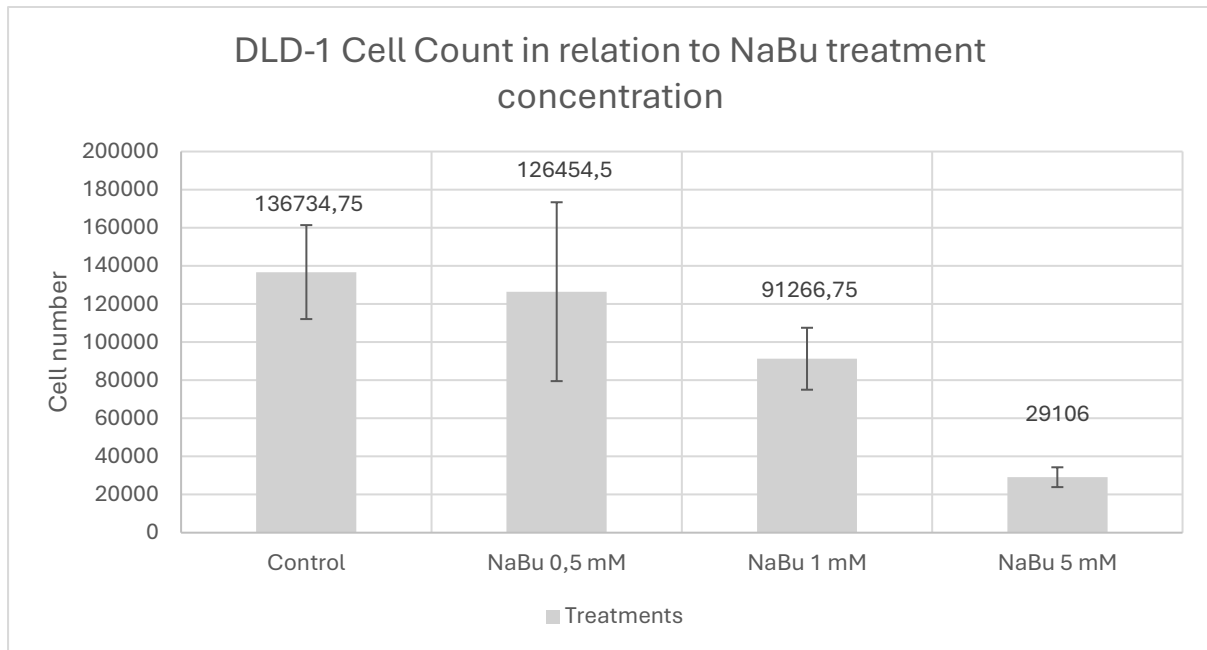


Figure 5. Effect of NaBu on DLD-1 cell count. Mean cell count of DLD-1 cells following treatment with increasing concentration of NaBu (0,5–5 mM). Values represent the mean $\pm$ standard deviation of four technical replicates.

This was repeated with CaCO₂, giving us another visual on the concentration of NaBu on the cell count.



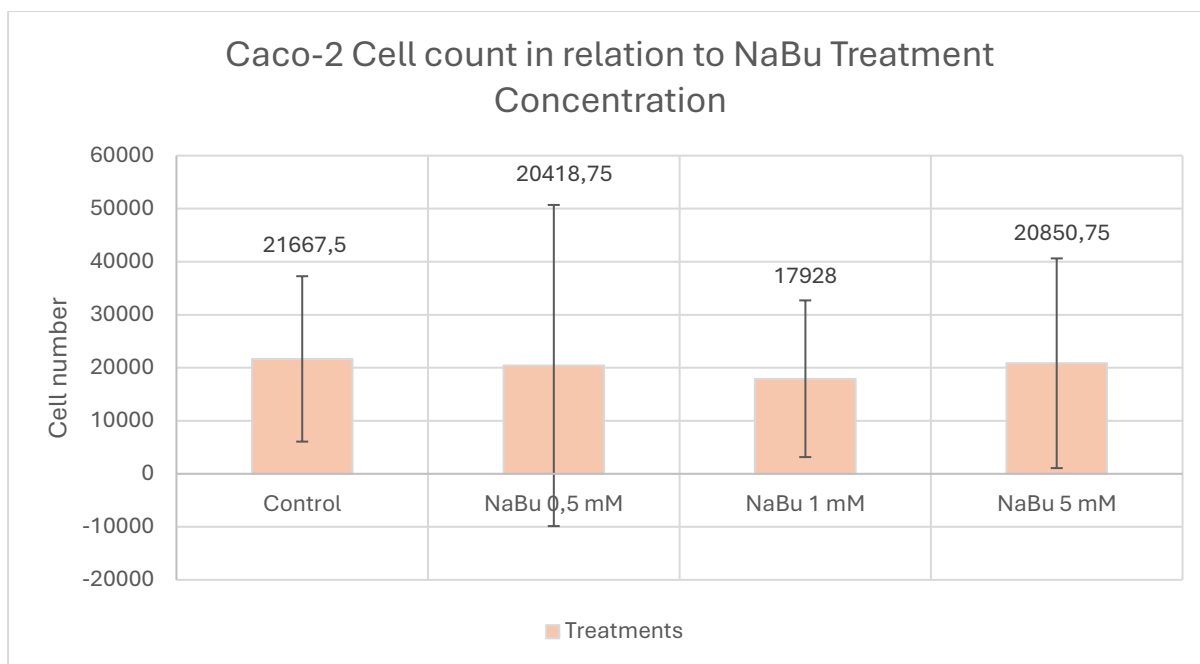


Figure 6. Effect of NaBu on Caco-2 cell count. Mean cell count of DLD-1 cells following treatment with increasing concentration of NaBu (0,5–5 mM). Values represent the mean $\pm$ standard deviation of four technical replicates.

For the second week of the experiment, the focus was put on one cell line, DLD-1 and the compound Resveratrol. For this part, the coulter count data for the two plates containing the different concentrations of Resveratrol, either with or without NaBu, was put through the same calculation template as the CaCO-2 and week 1 DLD-1. However, the cell count for the two plates were merged when calculating the mean and standard deviation to create a unified plot (Figure 7). We also divided the graphs between the NaBu treated and Non NaBu treat Cell culture at different resveratrol.



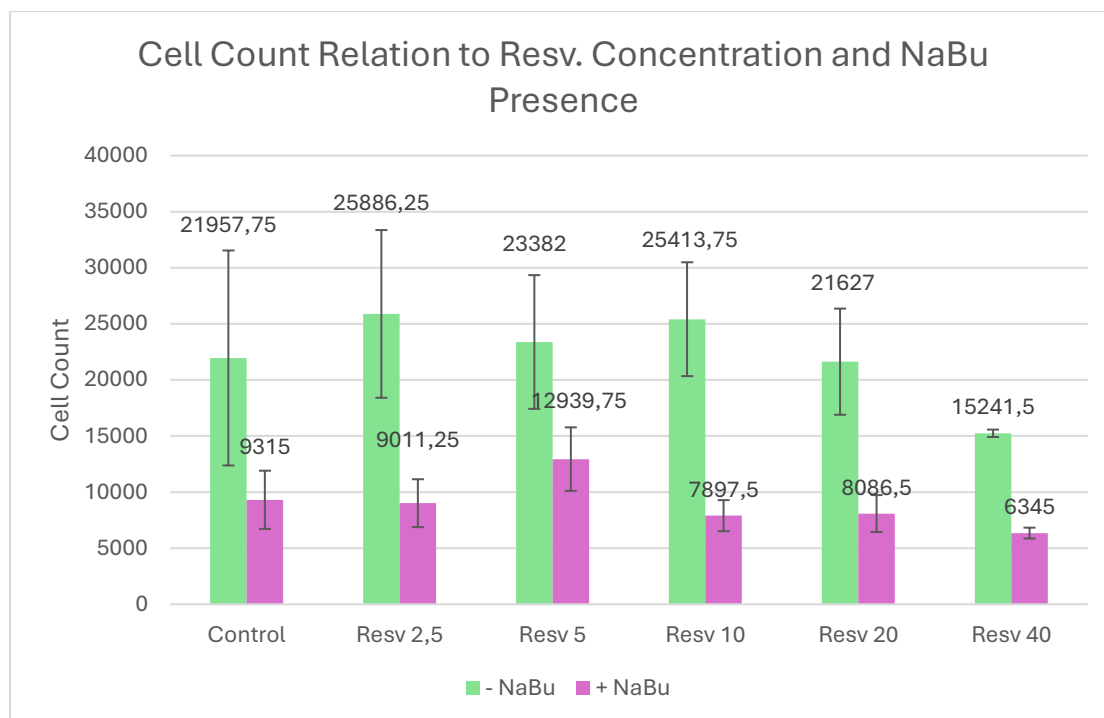


Figure 7. Effect of resveratrol on DLD-1 cell count with and without NaBu. Mean cell count of DLD-1 cells following treatment with increasing concentration of resveratrol (2.5–40 μM). Green represents cell count without NaBu. Red represents cell count with NaBu. Values represent the mean ± standard deviation of four technical replicates.

Cell area determination from phase contrast microscopy.

For the week 1 experiment, we see that the area of DLD-1 cells only shows a mild increase for increasing doses of NaBu, although the variance is also consistently higher than for the control cells (figure 8) 

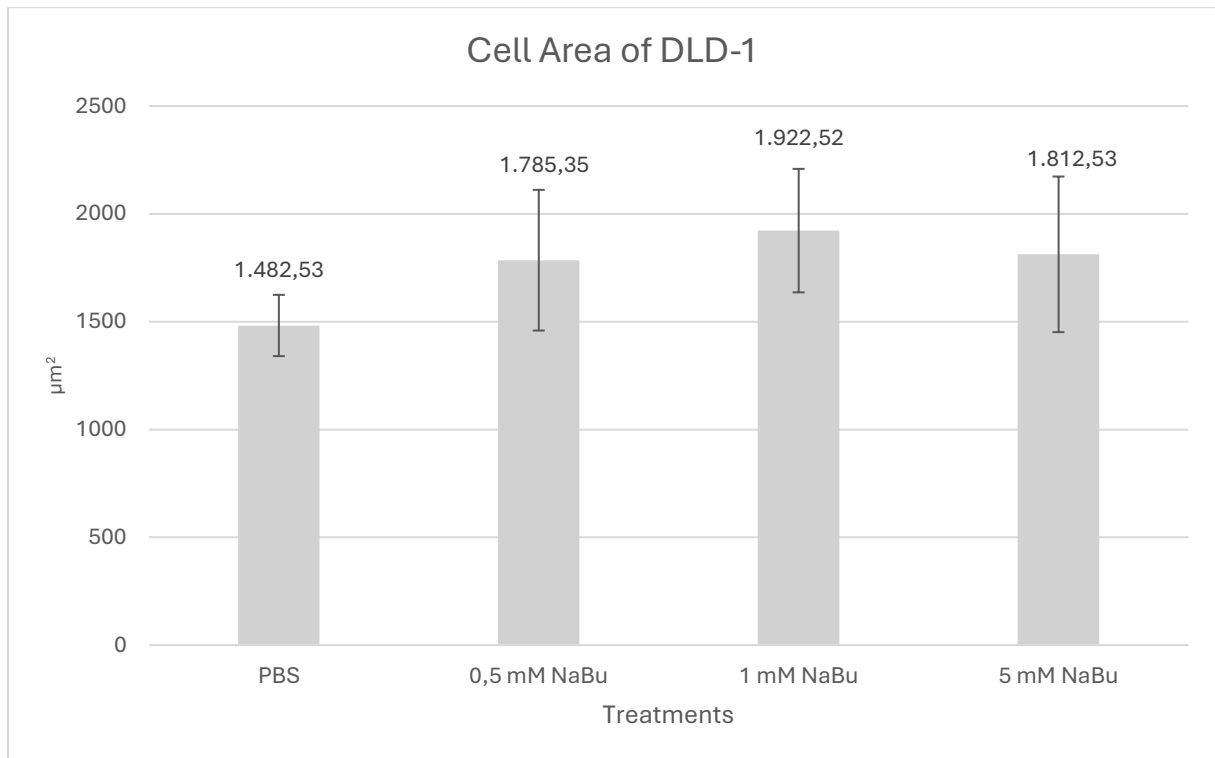


Figure 8. Effect of NaBu on DLD-1 cell area measured by ImageJ. Mean cell diameter in μm of Caco-2 cells after 48 h treatment with PBS (control), 0.5, 1 and 5 mM sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of three technical replicates.

The Caco-2 cells show a large increase for 0.5 mM NaBu and then smaller increases ones for the other doses. However, the high variance in this data set makes it complicated to say with any certainty whether there's a genuine trend here (figure 9).

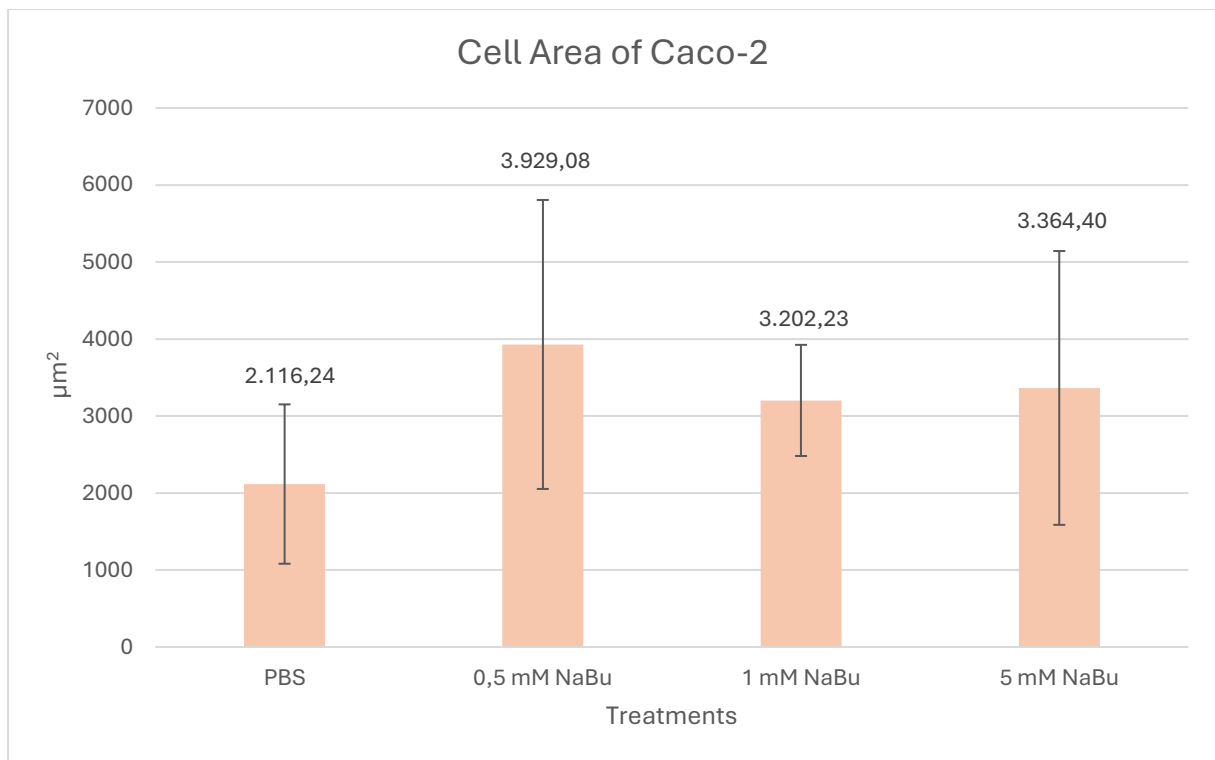


Figure 9. Effect of NaBu on Caco-2 cell area measured by ImageJ. Mean cell diameter in μm of Caco-2 cells after 48 h treatment with PBS (control), 0.5, 1 and 5 mM sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of three technical replicates.

For week 2 experiment, which only involved DLD-1 cells, we do see an increase in cell-size at the highest resveratrol dose (40 μM) while for the other doses it's about the same. Furthermore, the average cell area is consistently higher for the cells that were additionally treated with 1 mM NaBu. We again note that high variances here could potentially obscure biological patterns. 

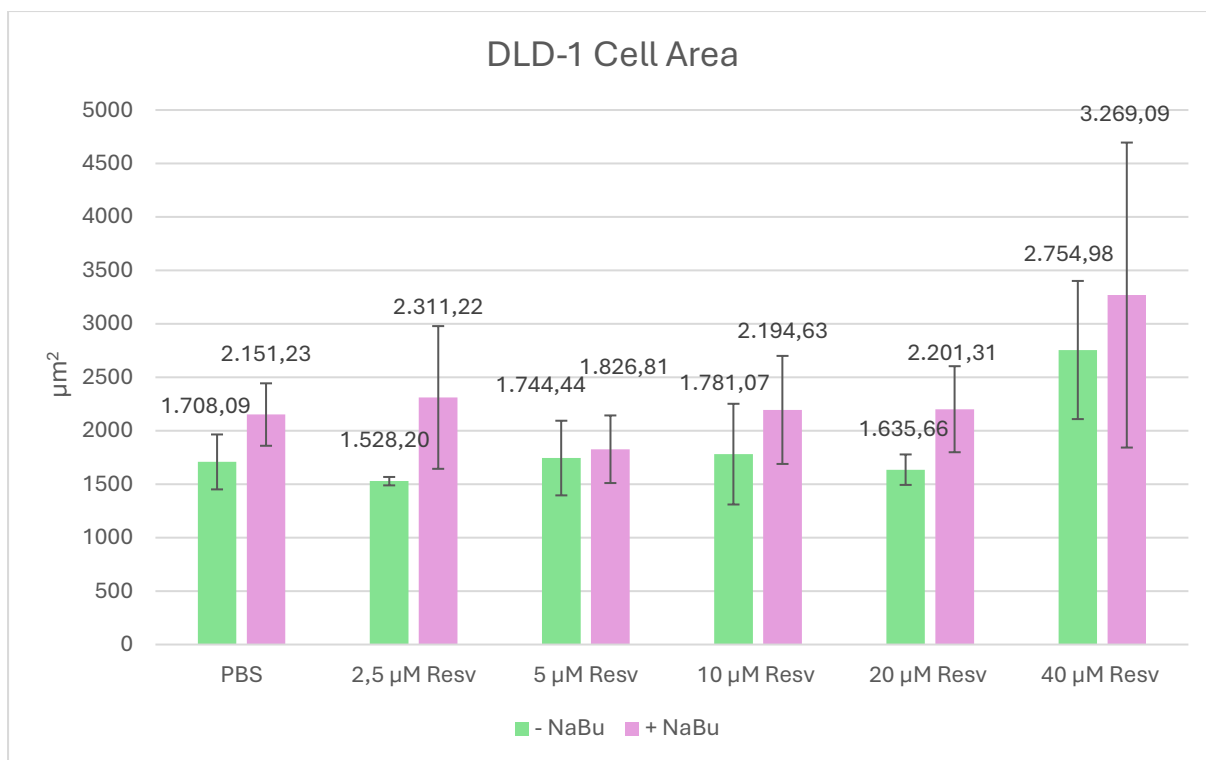


Figure 10. Effect of resveratrol with and without NaBu on DLD-1 cell areas measured by **ImageJ**. Mean cell diameter in μm of DLD-1 cells following treatment with increasing concentrations of resveratrol (2.5–40 μM) in the absence (green) or presence (purple) of sodium butyrate (NaBu). Values represent the mean $\pm$ standard deviation of four technical replicates.

Cellular structures

Figure 11 display phase contrast microscopy of DLD-1 cells treated only with PBS, we think that we can locate nucleus in most of the cells as a lighter oval region inside the cells. But we a limited in locating other cellular compartments, without the use of fluorescence staining.

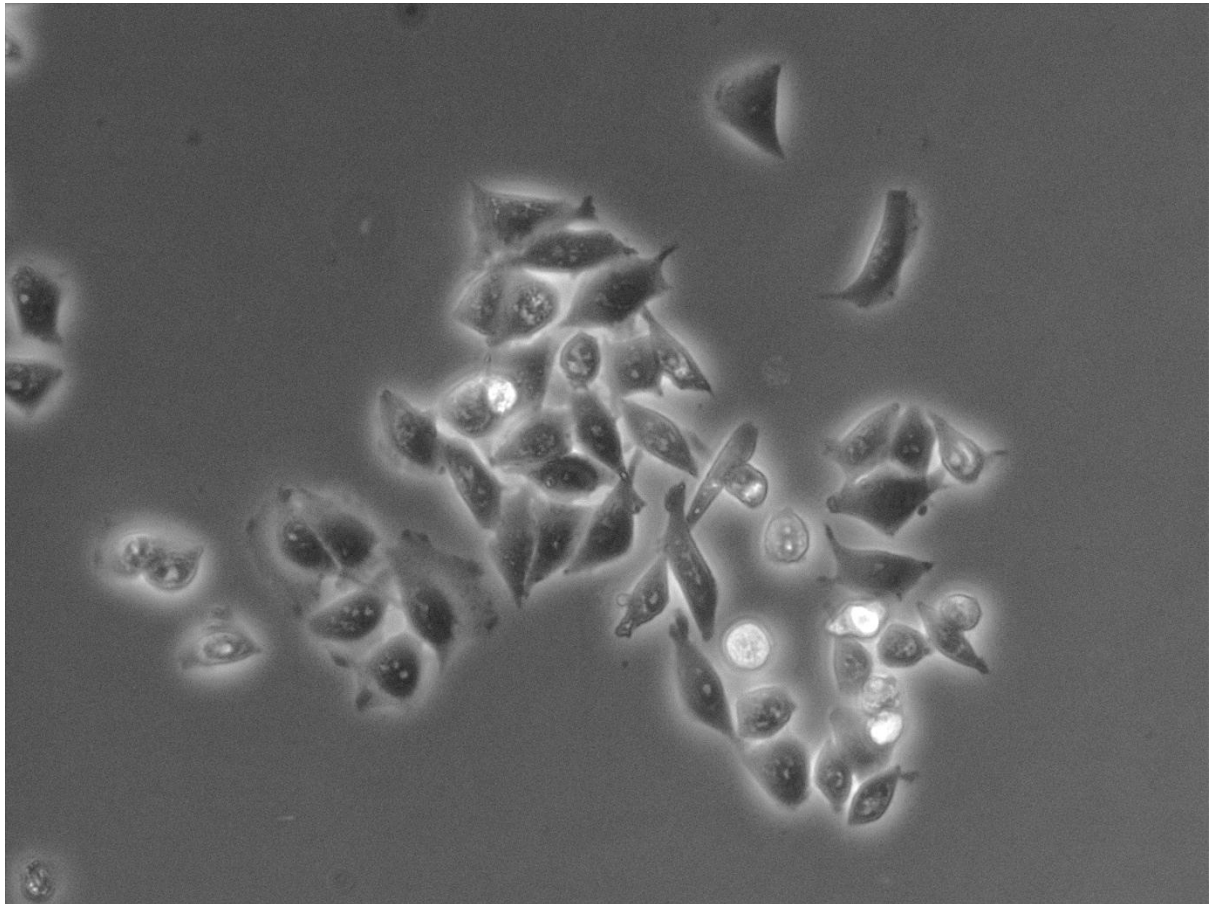


Figure 11: cellular structures. *The image has been captured using phase contrast microscopy 20× magnification, and further brighten 40% to better see cellular structures.*

Discussion

After gathering the data and plotting into graphs, patterns can be seen between resveratrol and NaBu on the cultures. Most changes was seen in DLD-1 as it was the focus of our experiment, with dedication of both our lab weeks to it. When looking at the figures, CaCO-2 also showed the least amount of change when treated with NaBu compared to DLD-1. When looking at the cell morphology and the cell count, DLD-1 seemed to be affected the most by the different concentrations. This suggests that NaBu and resveratrol impacts the morphological and proliferative abilities of the DLD-1 to a greater extent than it does on CaCO-2. This could mean that NaBu has cytotoxic effect, however, even though signs of dead cells were observed looking through the Phase-contrast microscope, the data collected can only tell us about the cells proliferative abilities rather than its cytotoxicity and thus cannot be confirmed.

The effects on the cells proliferative abilities can be seen with the increase in cell diameters combined with the decrease in cell count. This becomes more severe with the increase in the concentration of each of the compounds. NaBu, shows the strongest effect, with the effect of NaBu being immediately visible according to data. In comparison, you cannot see the immediate effect of resveratrol as we see with NaBu. Looking at figure 7 as an example, this is quite apparent with the cell count appearing to increase slightly between the 5 uM Resveratrol concentration and the 10 uM resveratrol concentration. Only when getting to the 20 and 40 uM Resveratrol concentration do we seen an effect on cell count.

Cell count

Increased NaBu concentration resulted in fewer DLD-1 cells after 48 hours of treatment, indicating that NaBu reduces the ability of DLD-1 cells to proliferate. Consistent with the increase in cell diameter observed by coulter counter measurements, indicating that the cells became larger while dividing less frequently. Also consistent with data from our cell-cycle from report 2, indicating a cell-cycle arrest or a cell-cycle accumulation in s-phase after NaBu-treatment.

The Caco-2 cells displayed modest changes in cell count following NaBu treatment. Suggesting caco-2 cells are less sensitive to NaBu than DLD-1 cells, under the experimental condition.

Cell area

When treating DLD-1 cells with NaBu we only see a modest increase in diameter as measured by the coulter counter data and the same is seen when quantifying area from phase contrast images. This is contrasted with the strong reduction in the number of cells as seen in the coulter counter experiment. Interestingly, Caco-2 cells appear to be much more unphased both in terms of cell counts and cell area. This indicates that certain metabolic or signalling pathways might be up or down-regulated in the cell-lines that cause them to respond very differently to these small metabolites, which is something researchers should take into account when trying to quantify the effect of various metabolites on certain cell-types, as both cell-lines in this case are of colorectal origin.

Similarly, the addition of resveratrol to DLD-1 cells causes a modest increase in cell-size based on our imaging data, which is mirrored by a modest decrease in the number of cells in our coulter counting data. Addition of 1 mM NaBu to the resveratrol treatment significantly reduces cell counts and induces a modest increase in cell size as quantified by phase-contrast microscopy.



Discussion of reproducible data in cell cultures

It can be challenging to generate reproducible data in cell culture experiments because of sensitivity to handling procedures and environments. Every person handles the cell culture differently, and the environment can also change. Even if you try to mimic, the cells can still be sensitive to any changes. Factors such as batches can change, and therefore also affect reproducibility. Several factors such as incomplete splitting and solutions not mixed well enough, or old reagents that have stopped working, can all act in different ways to produce heterogeneity in the data that might obscure true biological patterns or induce spurious correlations that aren't reproducible.

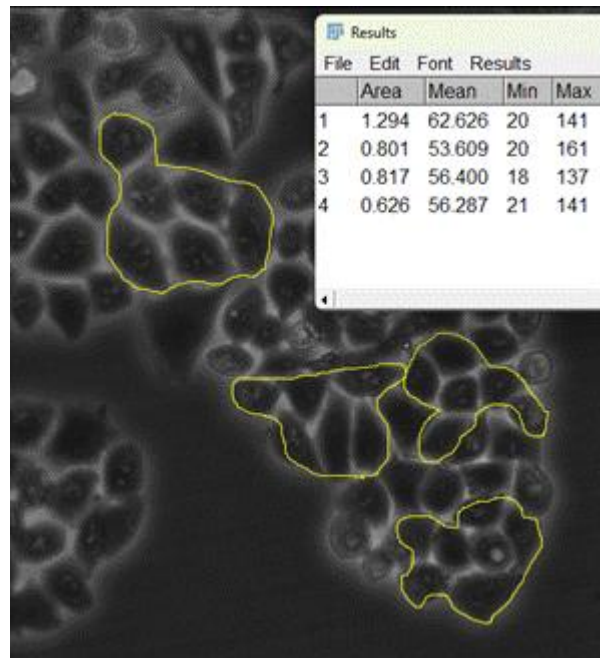


Appendix

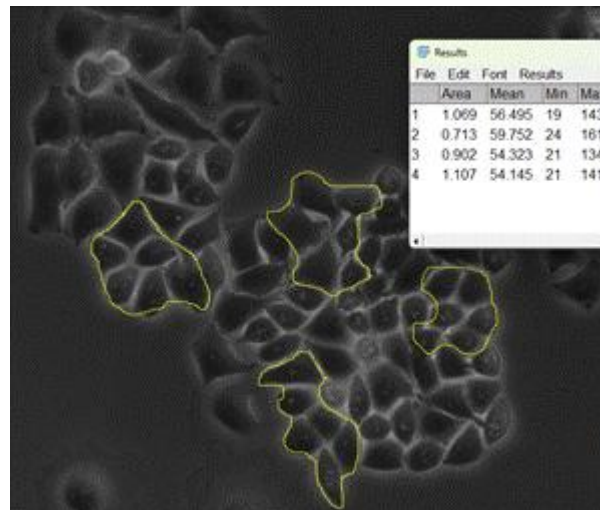
Week 1 DLD-1 Phase-Contrast

DLD-1

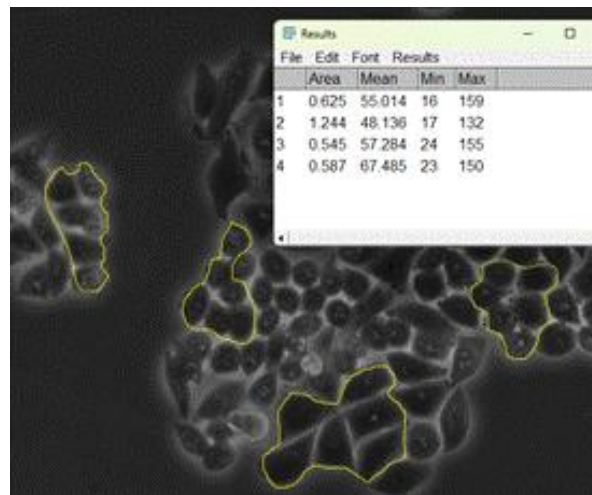
A1: Control
Six cells per
measurment



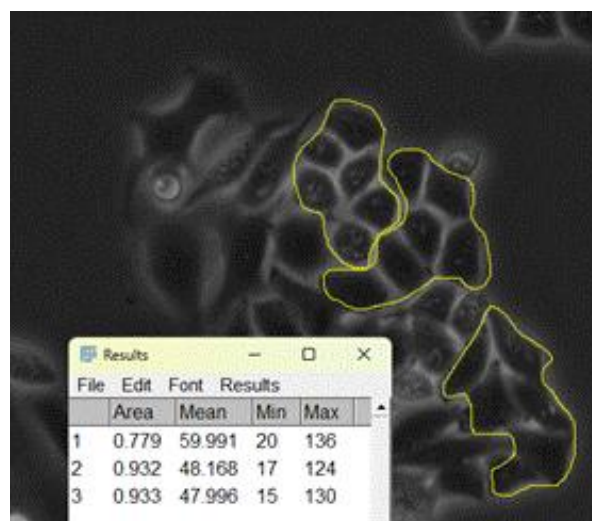
B1: Control
Six cells per
measurment



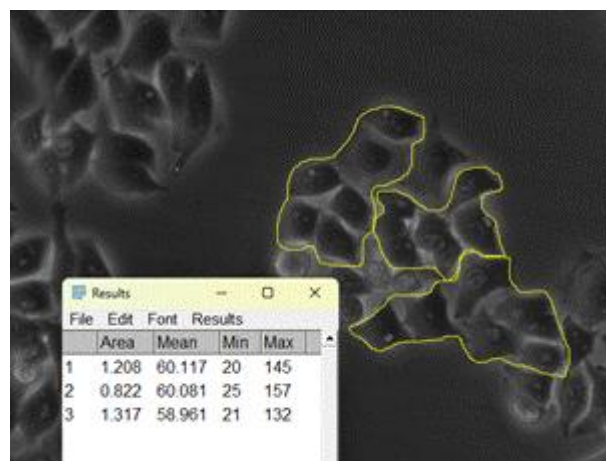
C1: Control
Six cells per
measurement



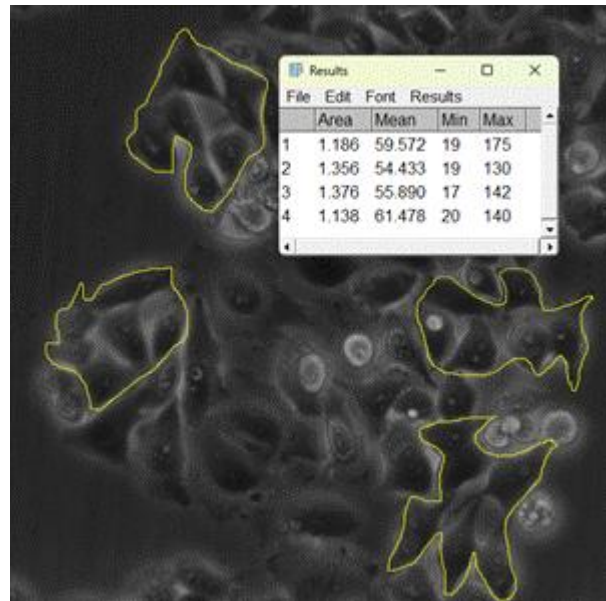
D1: Control
Six cells per
measurement



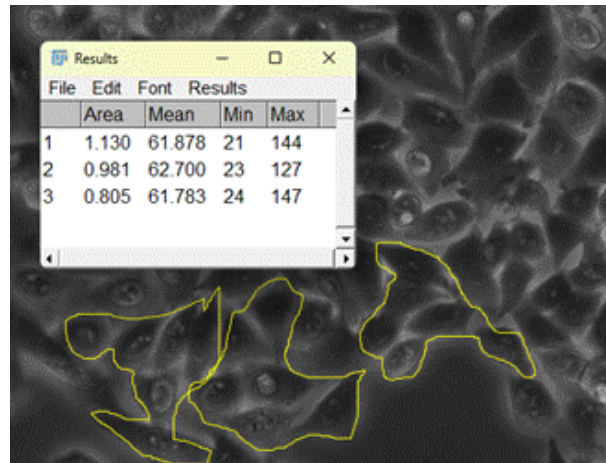
A2: 0.5 NaBu
Six cells per
measurement



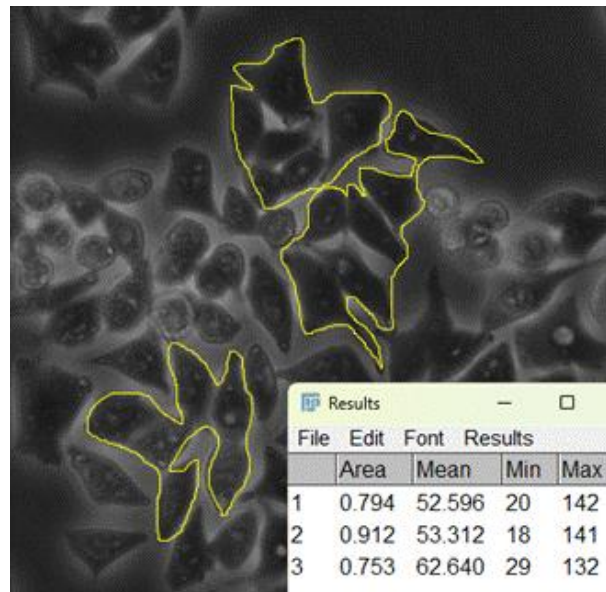
B2: 0.5 NaBu
Six cells per
measurement



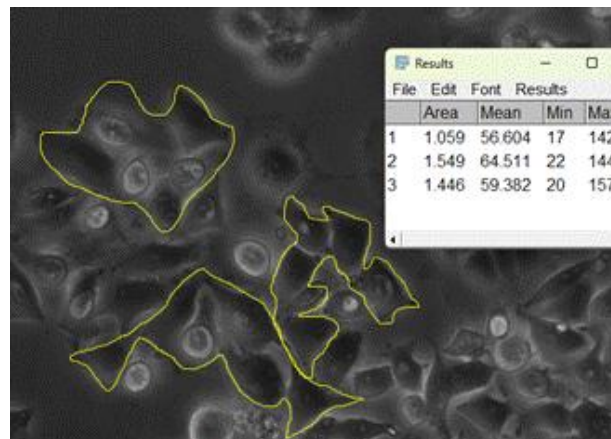
C2: 0.5 NaBu
Six cells per
measurement



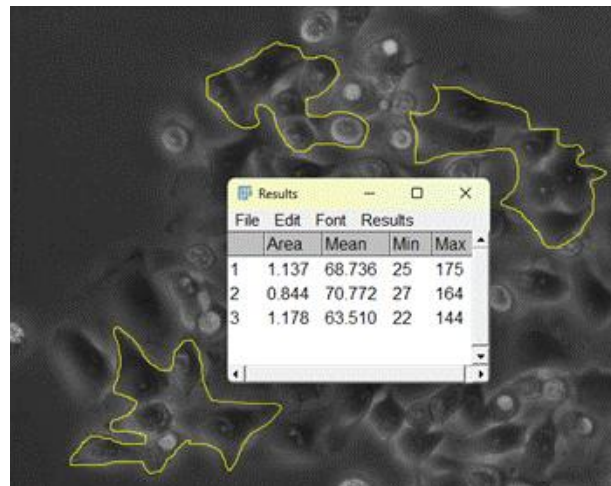
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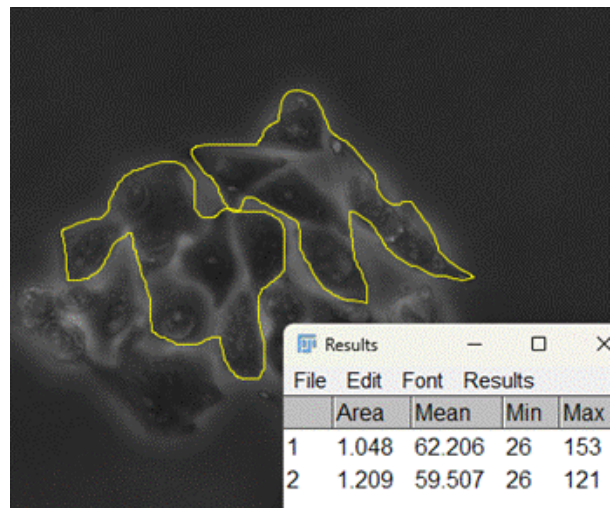
A3: 1 NaBu
Six cells per
measurement



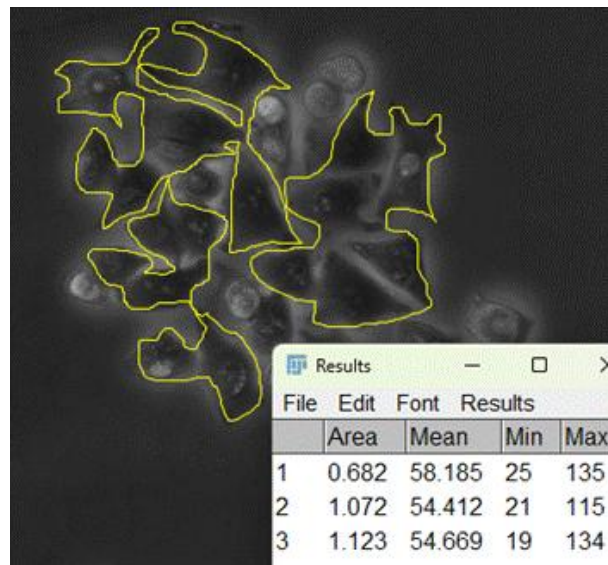
B3: 1 NaBu
Six cells per
measurement



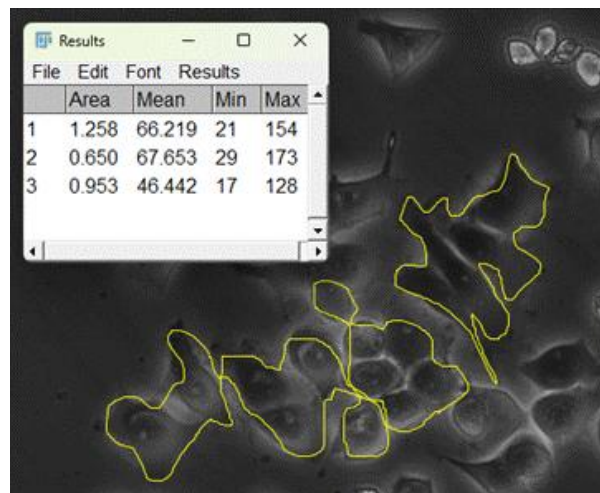
C3: 1 NaBu
Six cells per
measurement



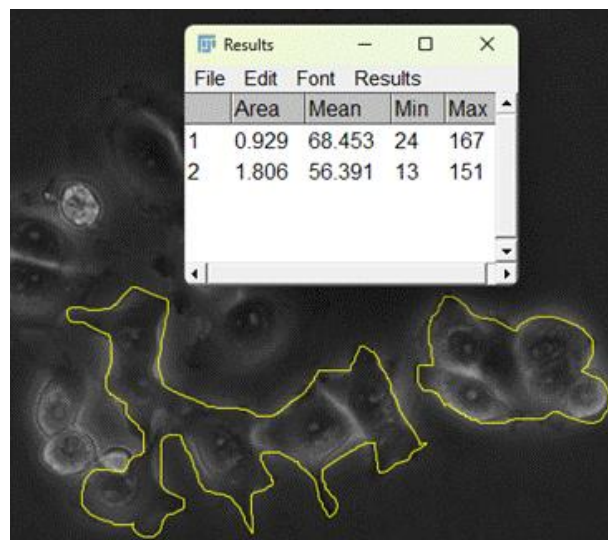
D3: 1 NaBu
Six cells per
measurement



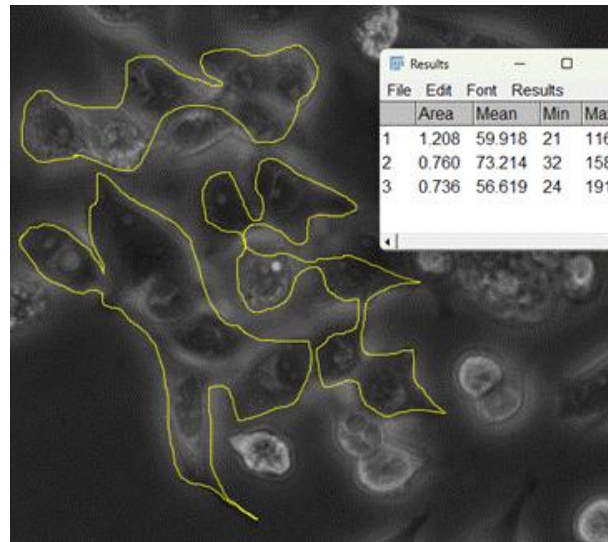
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Six cells per
measurement



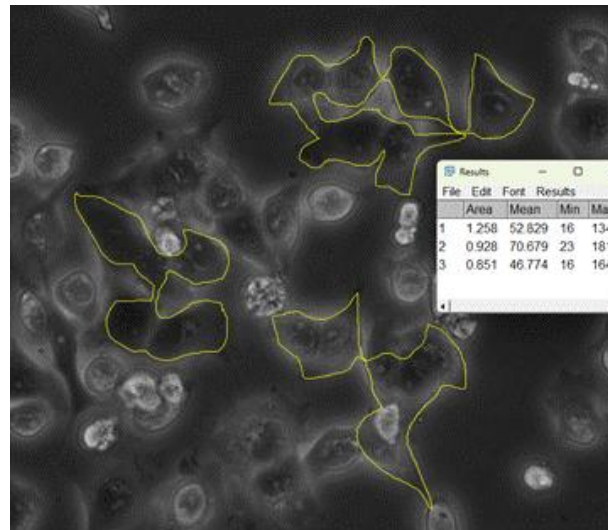
B4: 5 NaBu
Six cells per
measurement



C4: 5 NaBu
Six cells per
measurement



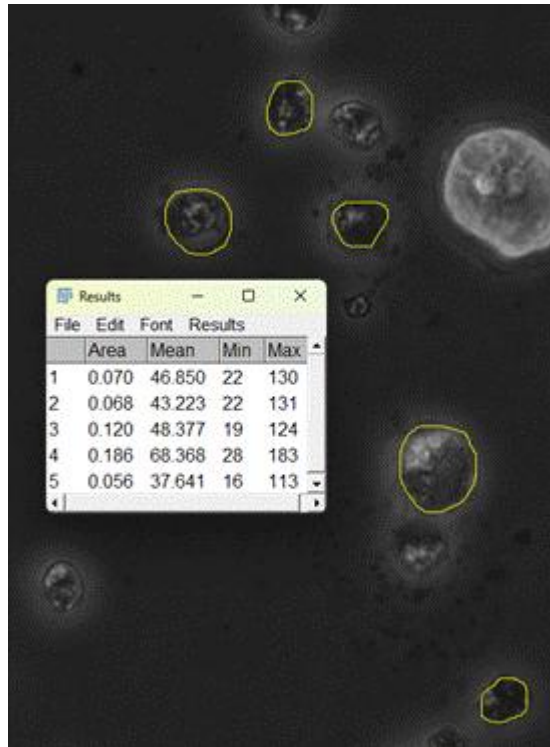
D4: 5 NaBu
Six cells per
measurement



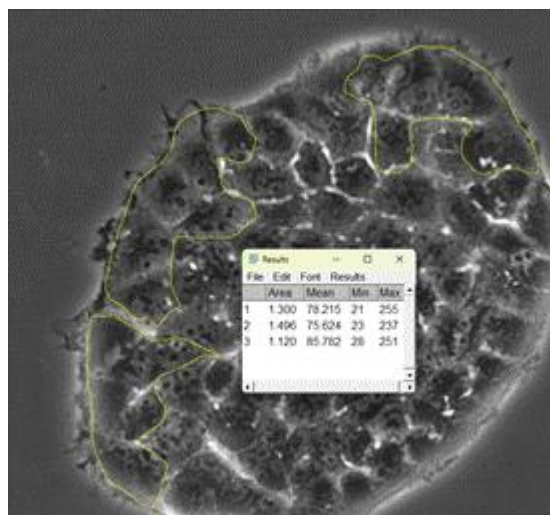
Week 1 CaCO-2:

CaCO-2:

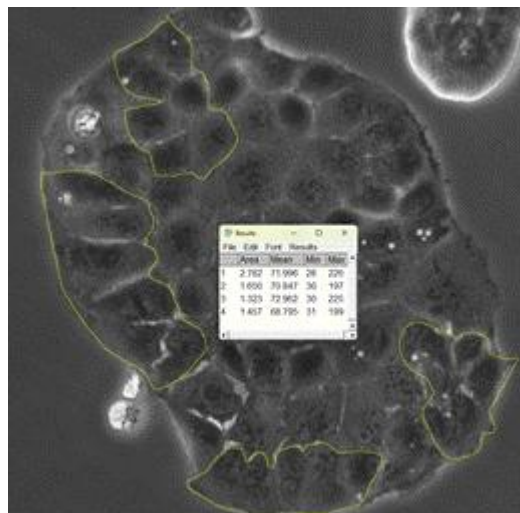
A1: Control



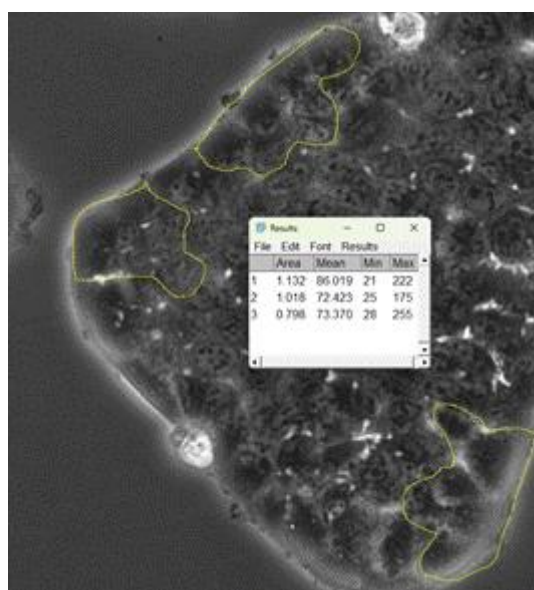
B1: Control
Six cells per
measurement



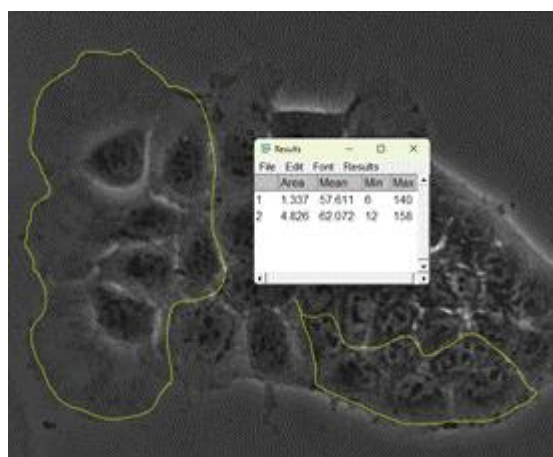
C1: Control
Six cells per
measurement



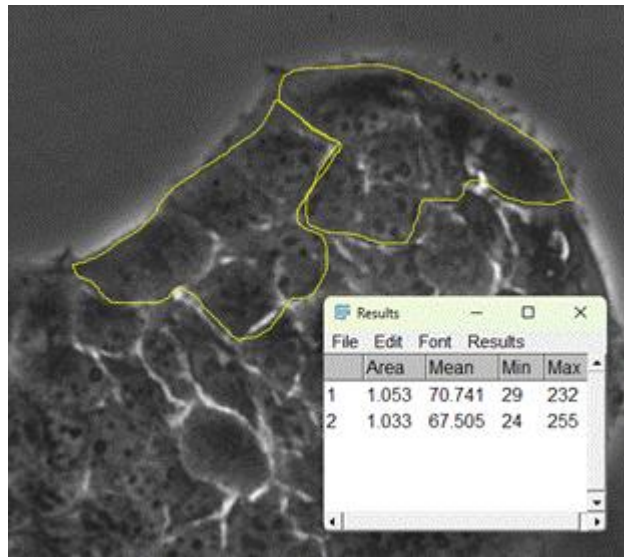
D1: Control



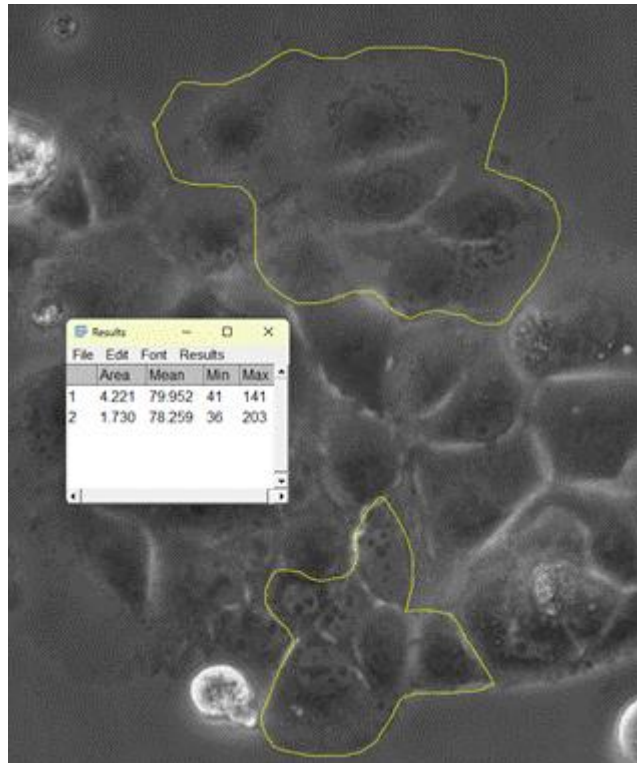
A2: 0.5 NaBu
Six cells per
measurement



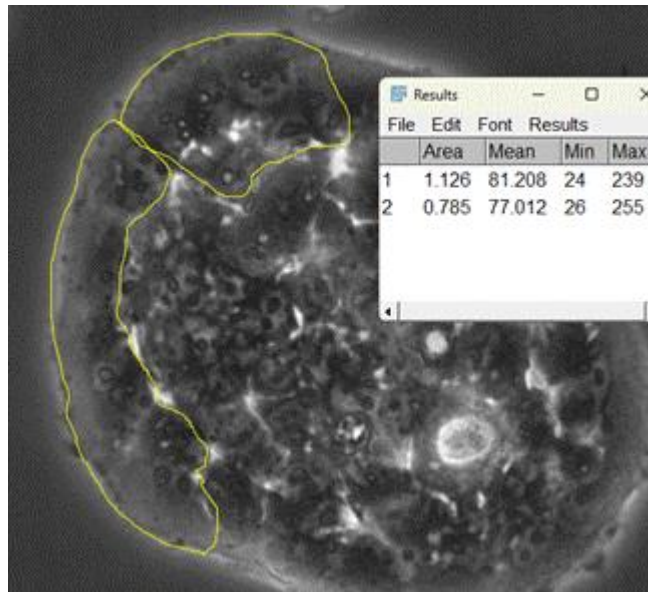
B2: 0.5 NaBu
Six cells per
measurement



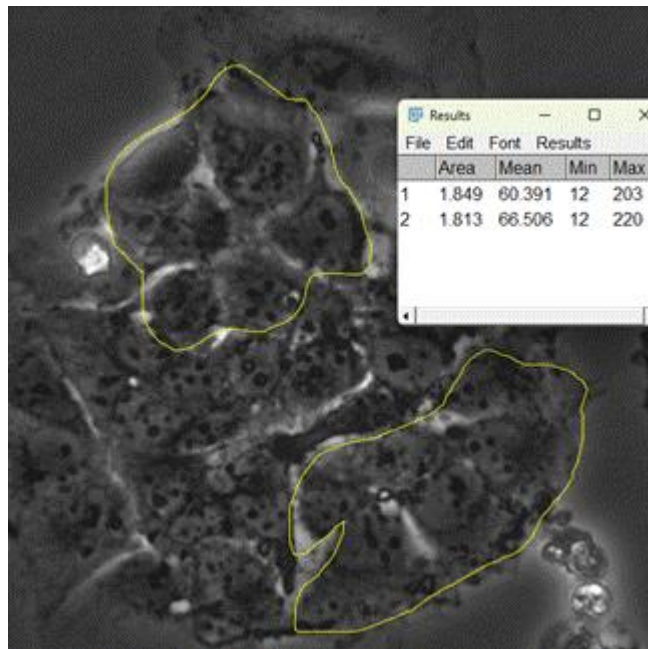
C2: 0.5 NaBu
Six cells per
measurement



D2: 0.5 NaBu
Six cells per
measurement



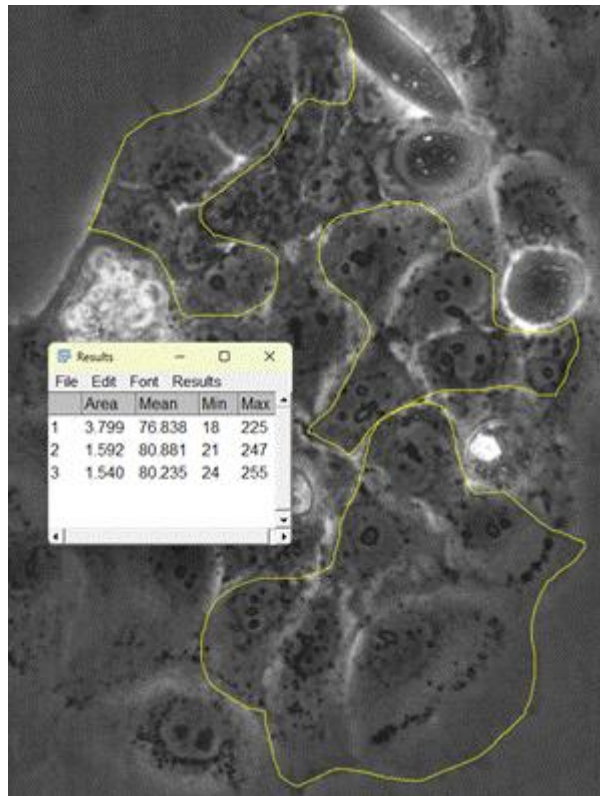
A3: 1 NaBu
Six cells per
measurement



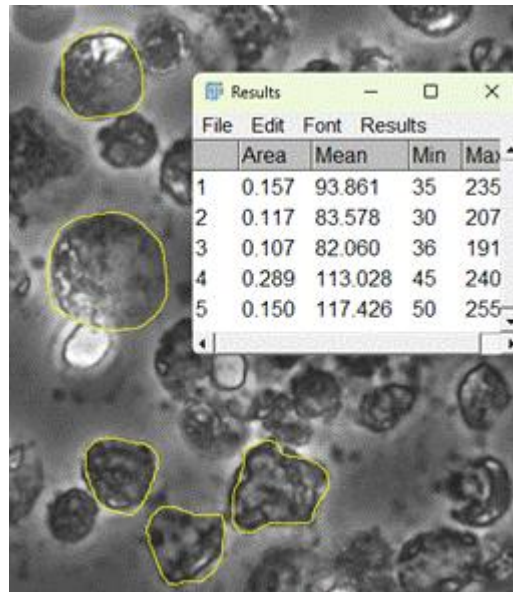
B3: 1 NaBu
Six cells per
measurement



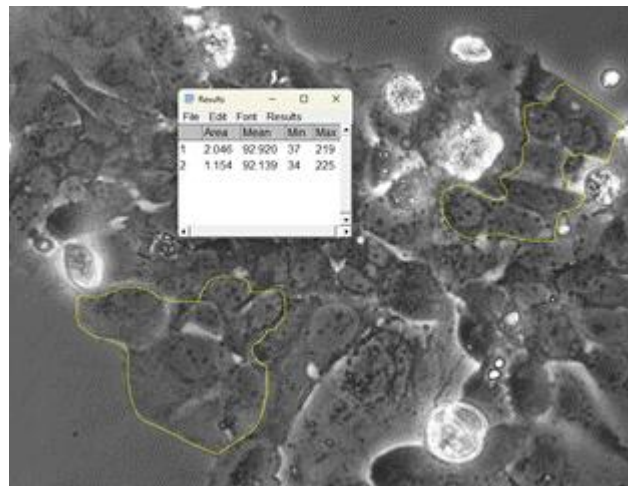
C3: 1 NaBu
Six cells per
measurement



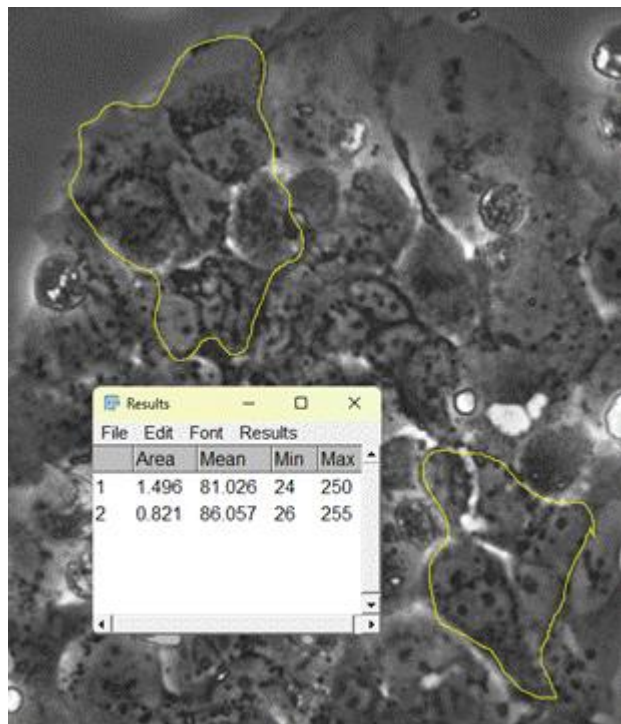
D3: 1 NaBu



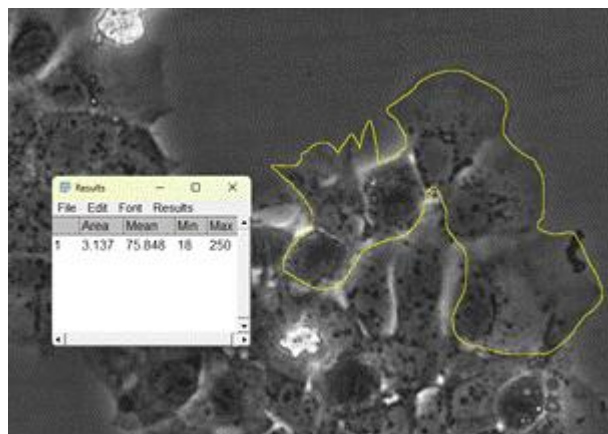
A4: 5 NaBu
Six cells per
measurement



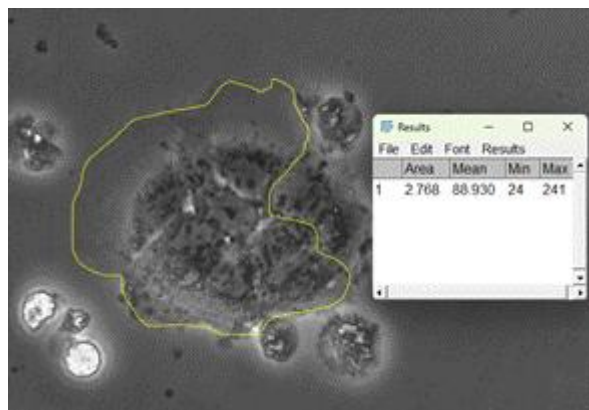
B4: 5 NaBu
Six cells per
measurement



C4: 5 NaBu
Six cells per
measurement

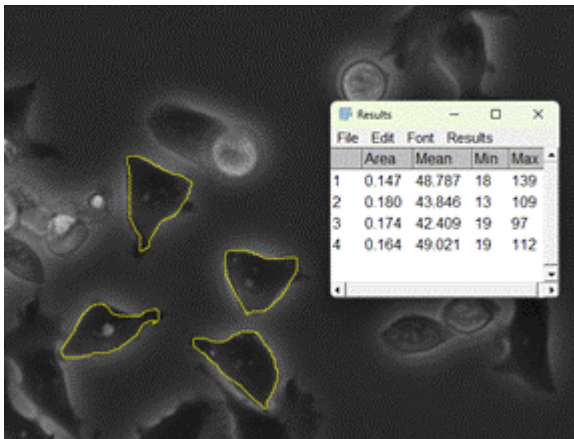
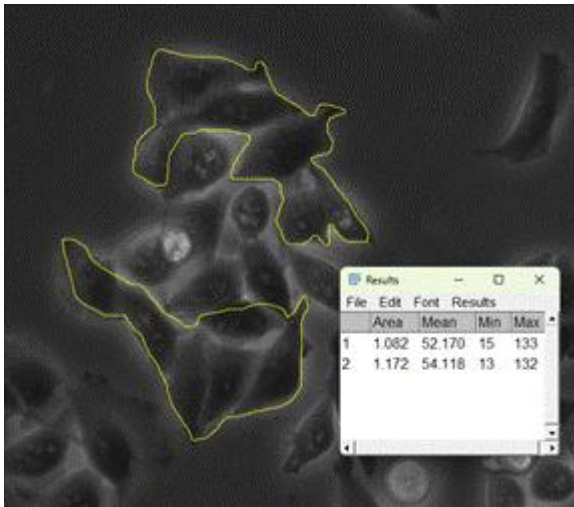
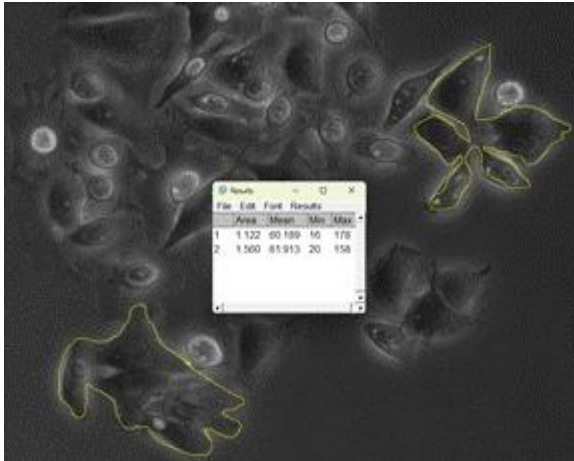


D4: 5 NaBu
Six cells per
measurement

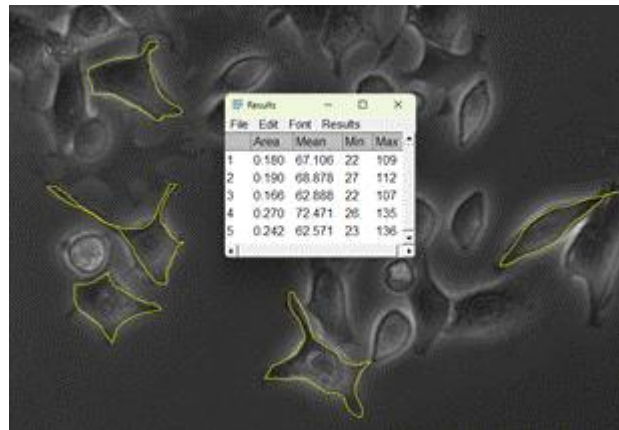


Week 2 DLD-1 Plate 1:

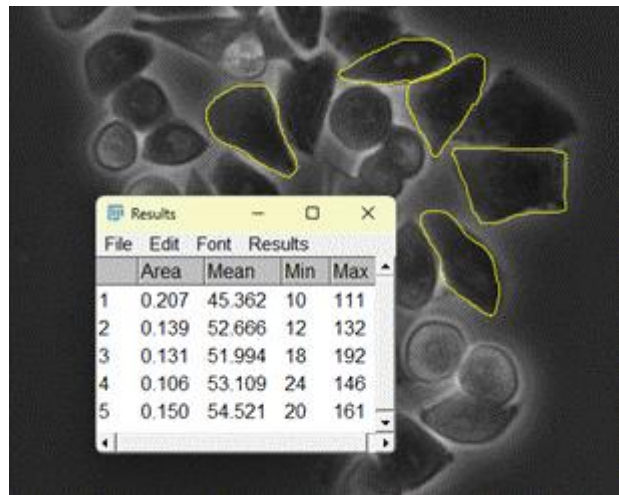
DLD-1: 1

A1: Control	 <table><tr><th></th><th>Area</th><th>Mean</th><th>Min</th><th>Max</th></tr><tr><td>1</td><td>0.147</td><td>48.787</td><td>18</td><td>139</td></tr><tr><td>2</td><td>0.180</td><td>43.846</td><td>13</td><td>109</td></tr><tr><td>3</td><td>0.174</td><td>42.409</td><td>19</td><td>97</td></tr><tr><td>4</td><td>0.164</td><td>49.021</td><td>19</td><td>112</td></tr></table>		Area	Mean	Min	Max	1	0.147	48.787	18	139	2	0.180	43.846	13	109	3	0.174	42.409	19	97	4	0.164	49.021	19	112
	Area	Mean	Min	Max																						
1	0.147	48.787	18	139																						
2	0.180	43.846	13	109																						
3	0.174	42.409	19	97																						
4	0.164	49.021	19	112																						
B1: Control Six cells per measurement	 <table><tr><th></th><th>Area</th><th>Mean</th><th>Min</th><th>Max</th></tr><tr><td>1</td><td>1.082</td><td>52.170</td><td>15</td><td>133</td></tr><tr><td>2</td><td>1.172</td><td>54.118</td><td>13</td><td>132</td></tr></table>		Area	Mean	Min	Max	1	1.082	52.170	15	133	2	1.172	54.118	13	132										
	Area	Mean	Min	Max																						
1	1.082	52.170	15	133																						
2	1.172	54.118	13	132																						
C1: NaBu Six cells per measurement	 <table><tr><th></th><th>Area</th><th>Mean</th><th>Min</th><th>Max</th></tr><tr><td>1</td><td>1.122</td><td>50.189</td><td>16</td><td>178</td></tr><tr><td>2</td><td>1.500</td><td>81.913</td><td>20</td><td>158</td></tr></table>		Area	Mean	Min	Max	1	1.122	50.189	16	178	2	1.500	81.913	20	158										
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1	1.122	50.189	16	178																						
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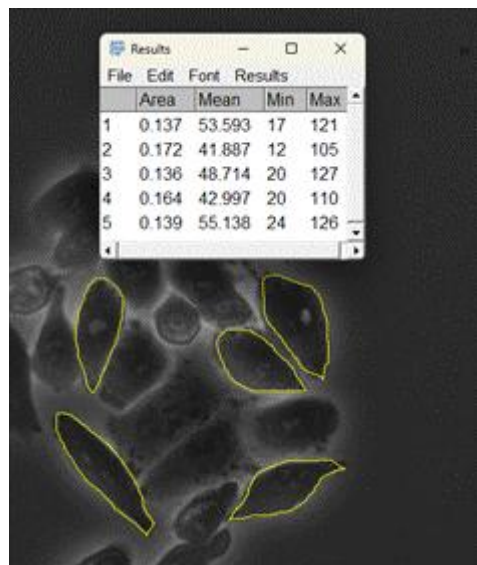
D1: NaBu



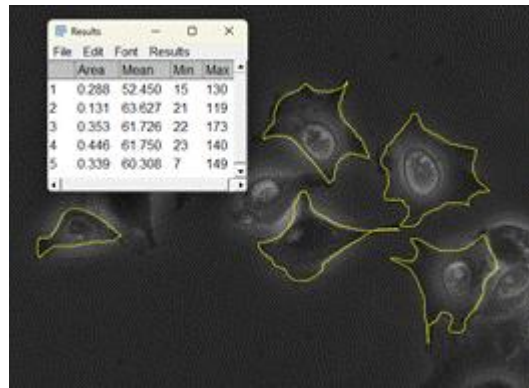
A2: 2,5 Resv



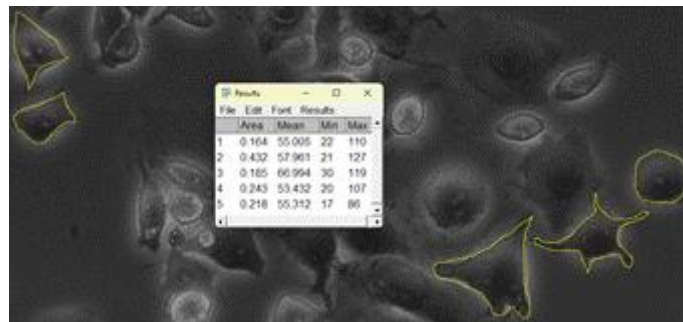
B2: 2,5 Resv



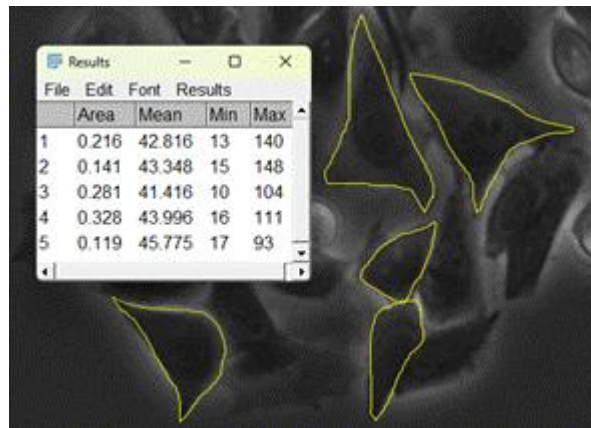
C2: 2,5 Resv +
NaBu



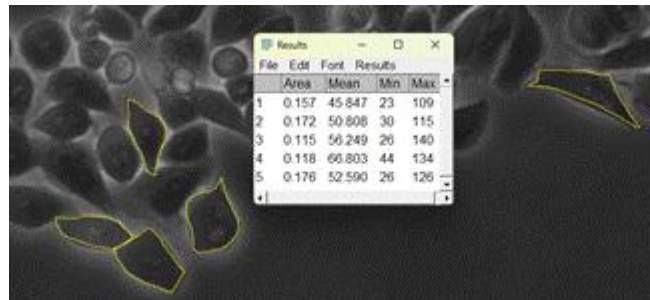
D2: 2,5 Resv +
NaBu



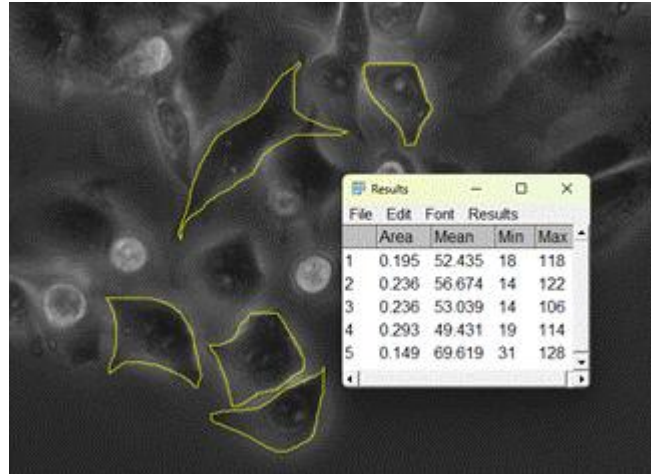
A3: 5 Resv



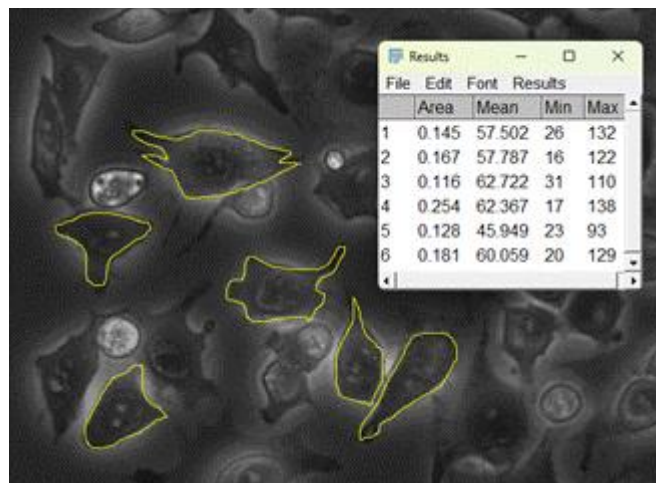
B3: 5 Resv



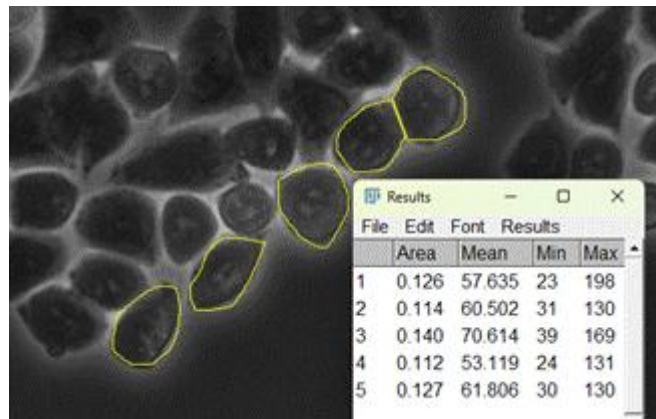
C3: 5 Resv +
NaBu



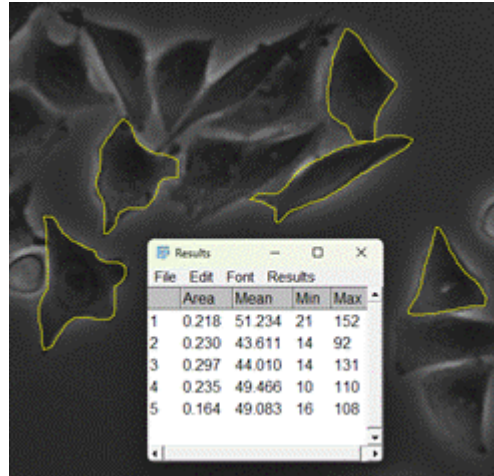
D3: 5 Resv +
NaBu



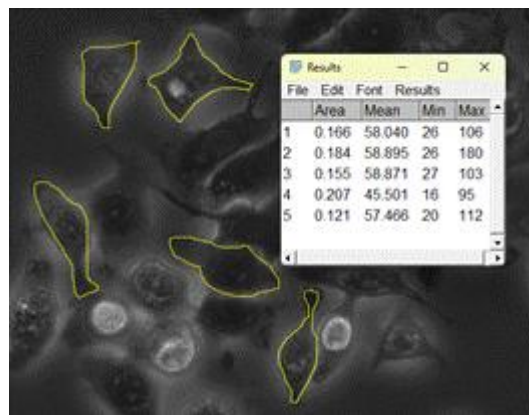
A4: 10 Resv



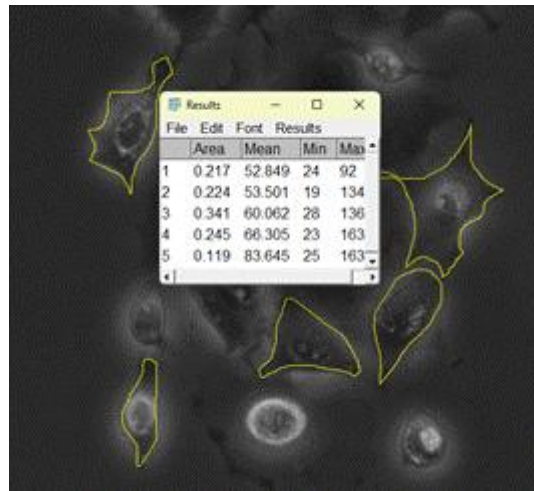
B4: 10 Resv



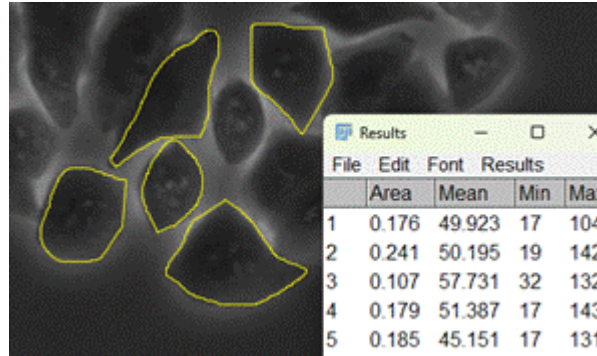
C4: 10 Resv + NaBu



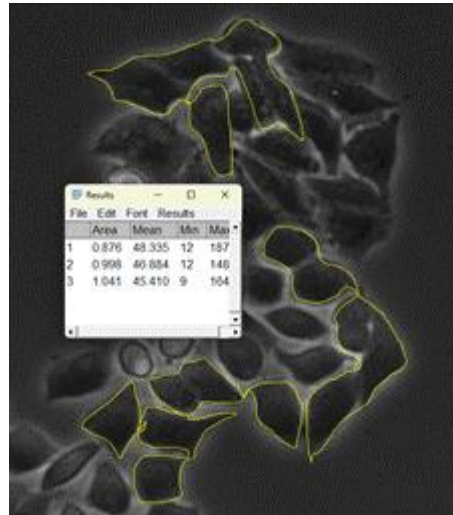
D4: 10 Resv + NaBu



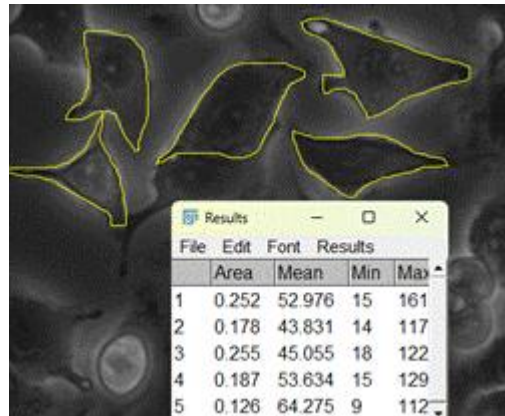
A5: 20 Resv



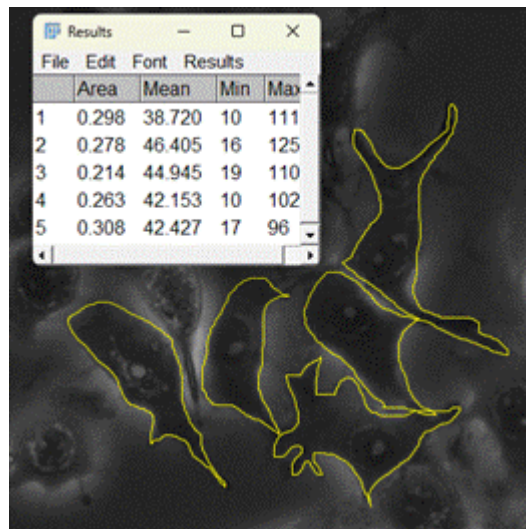
B5: 20 Resv



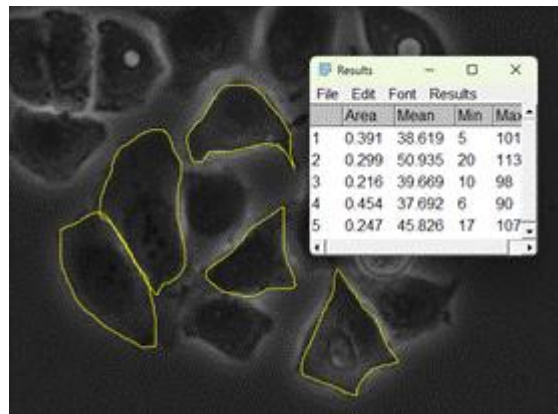
C5: 20 Resv + NaBu



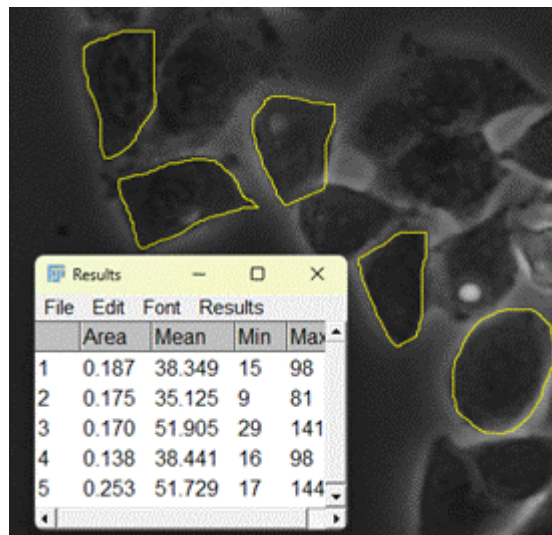
D5: 20 Resv +
NaBu



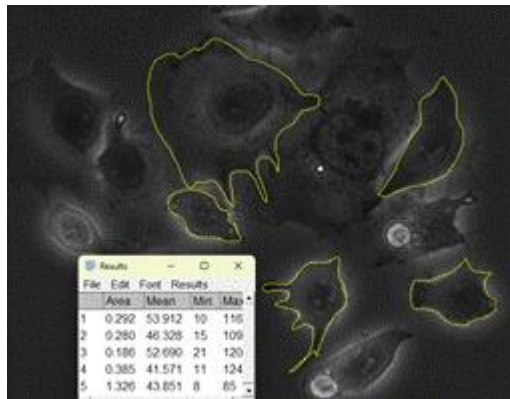
A6: 40 Resv



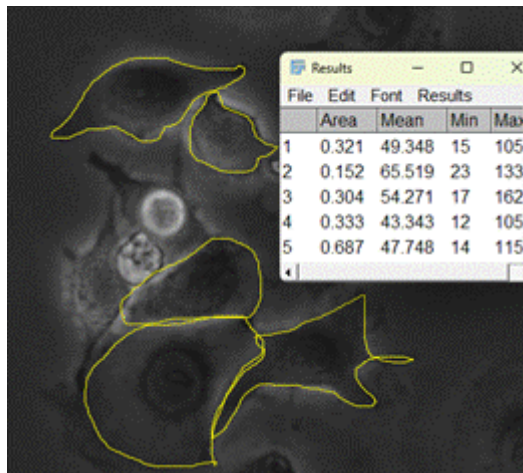
B6: 40 Resv



C6: 40 Resv +
NaBu

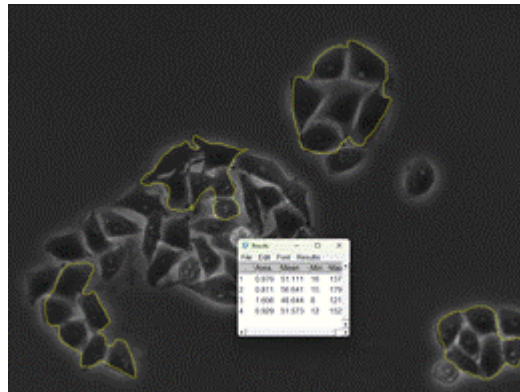


D6: 40 Resv +
NaBu

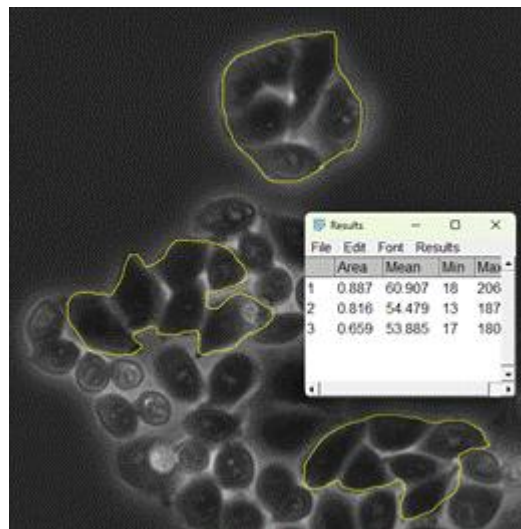


DLD-1 Week 2: plate 2

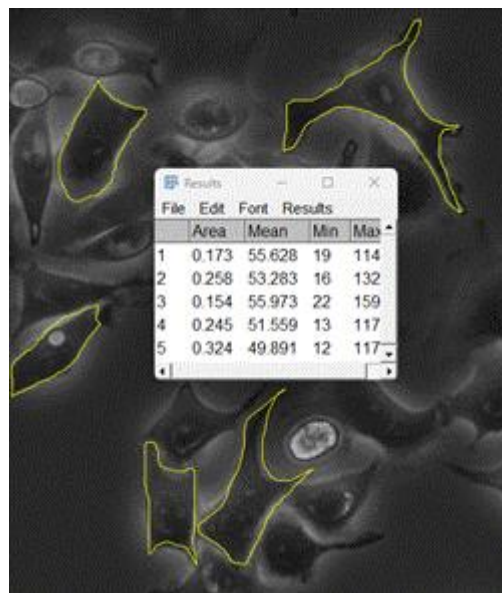
A1: Control



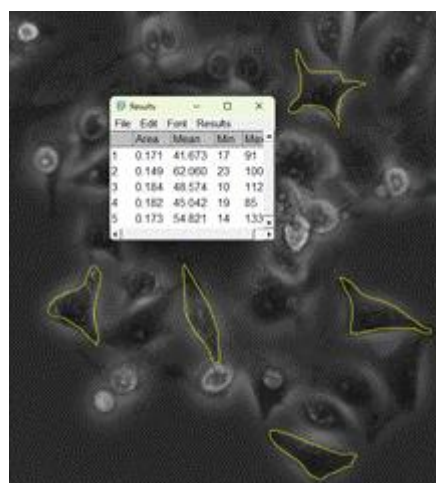
B1: Control
Six cells per
measurement



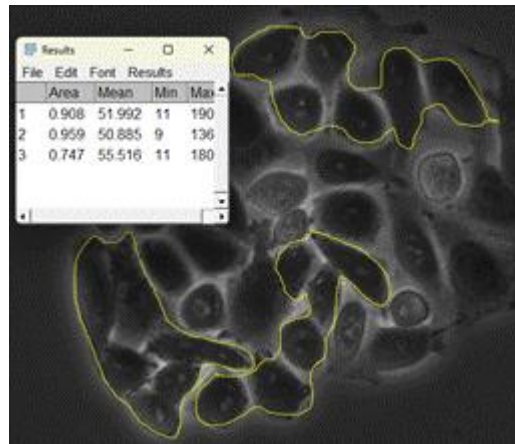
C1: NaBu
Six cells per
measurement



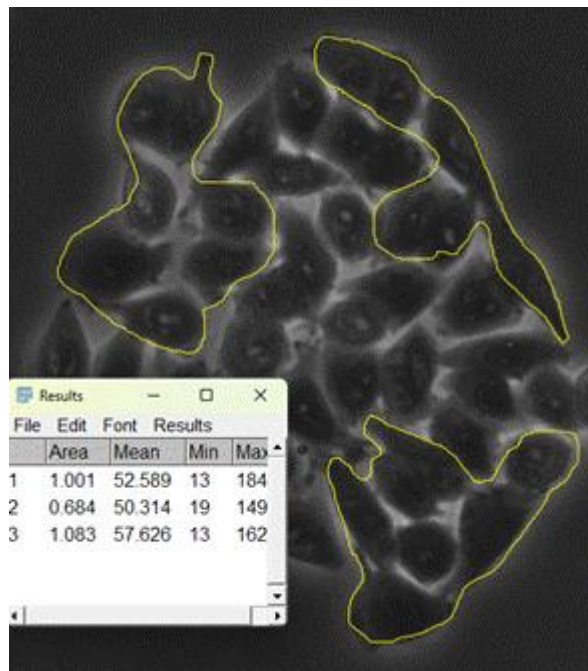
D1: NaBu



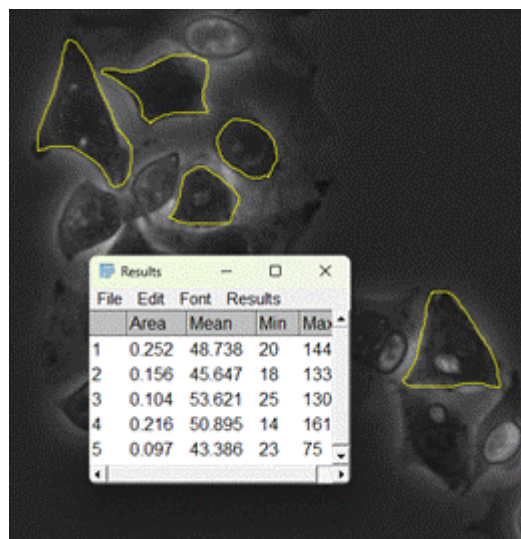
A2: 2,5 Resv



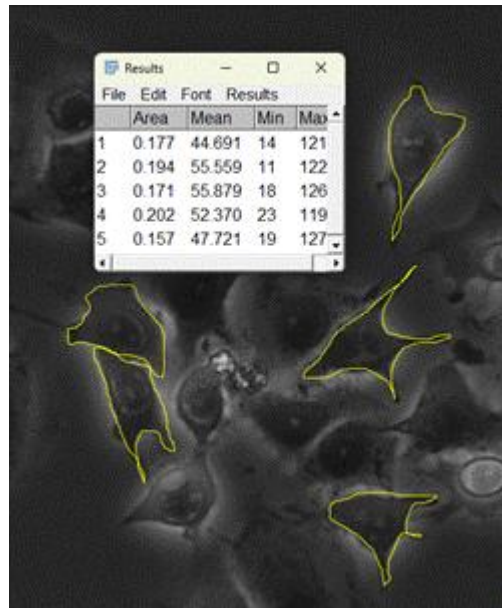
B2: 2,5 Resv



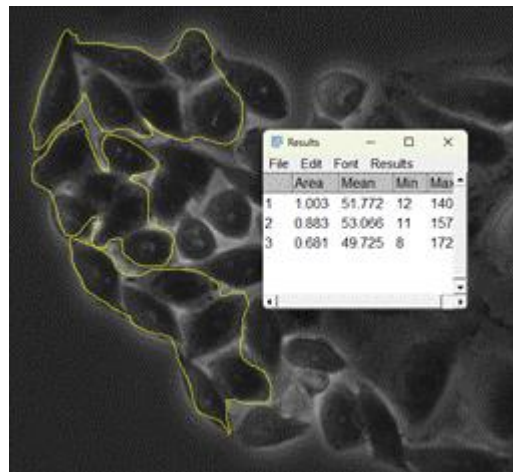
C2: 2,5 Resv +
NaBu



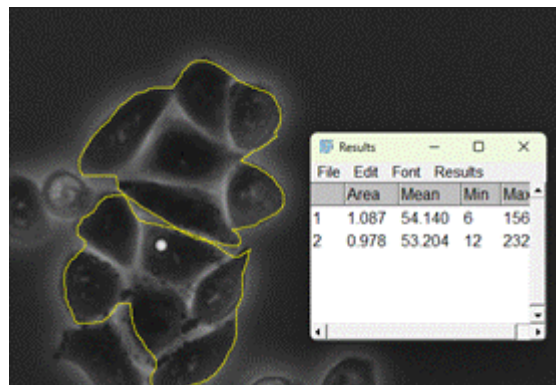
D2: 2,5 Resv +
NaBu



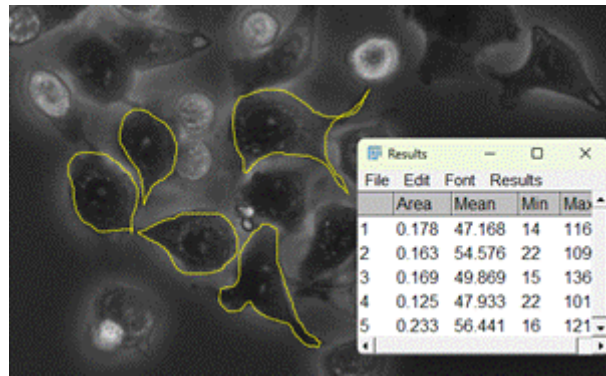
A3: 5 Resv



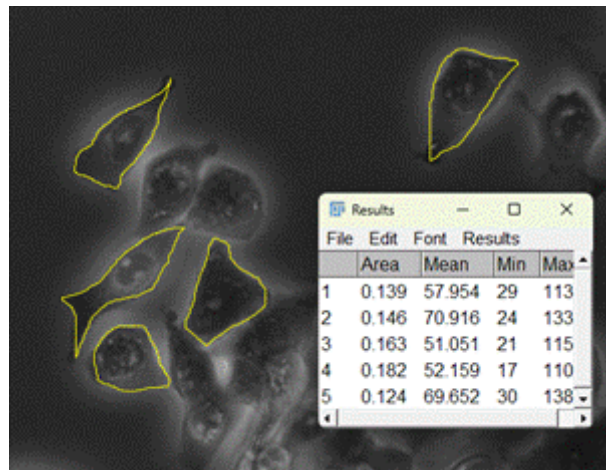
B3: 5 Resv



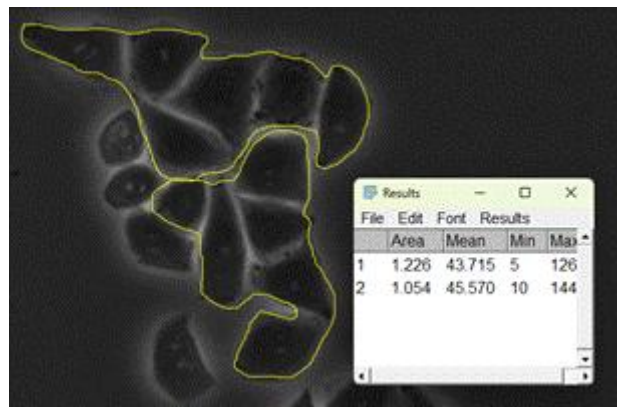
C3: 5 Resv + NaBu



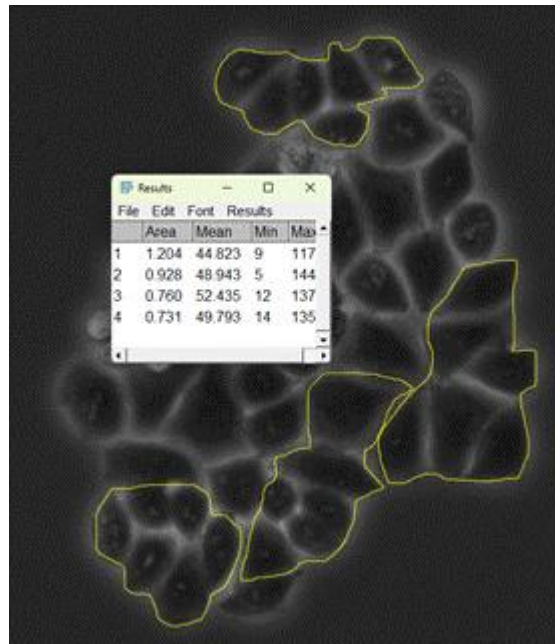
D3: 5 Resv + NaBu



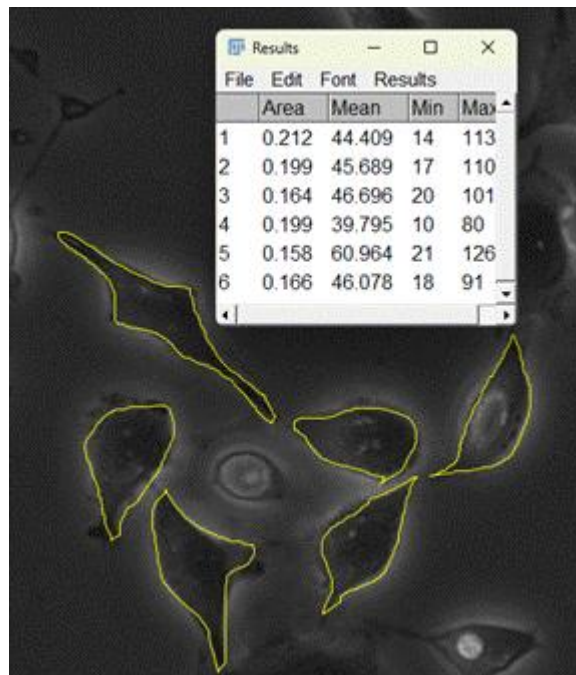
A4: 10 Resv



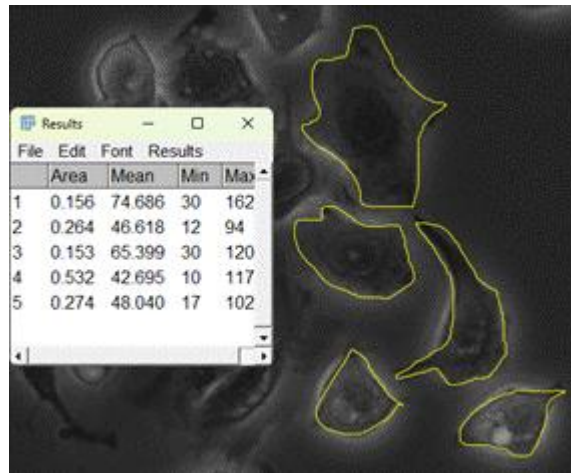
B4: 10 Resv



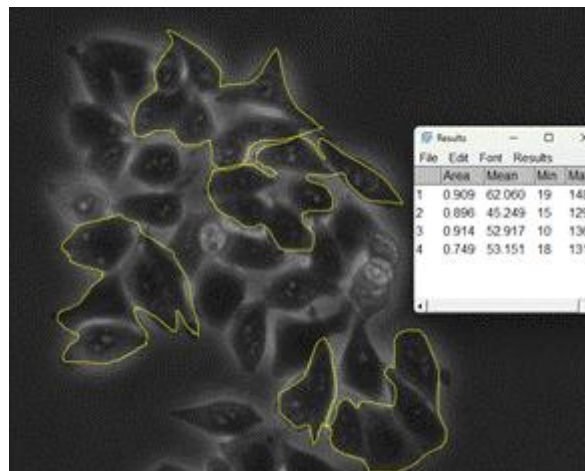
C4: 10 Resv +
NaBu



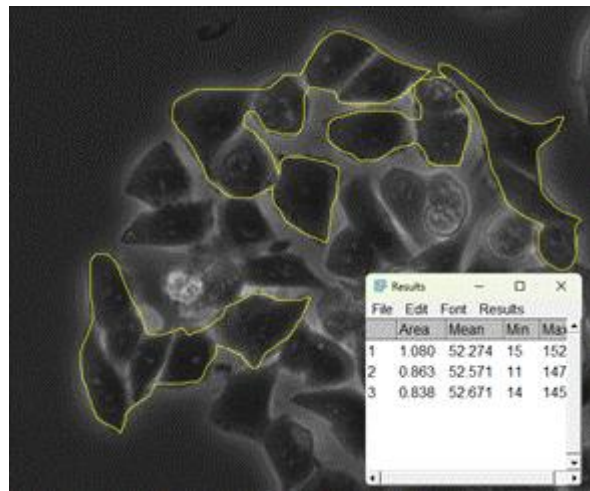
D4: 10 Resv +
NaBu



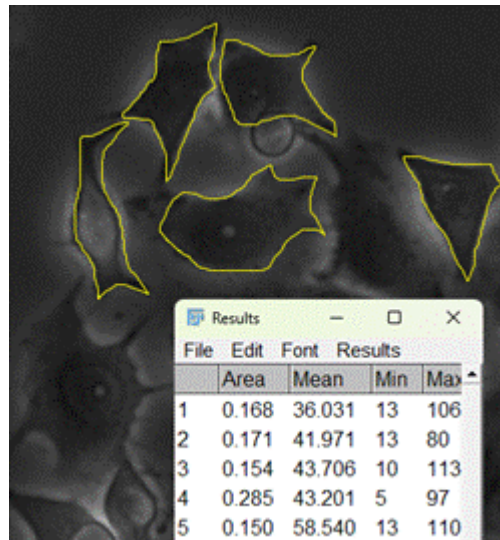
A5: 20 Resv



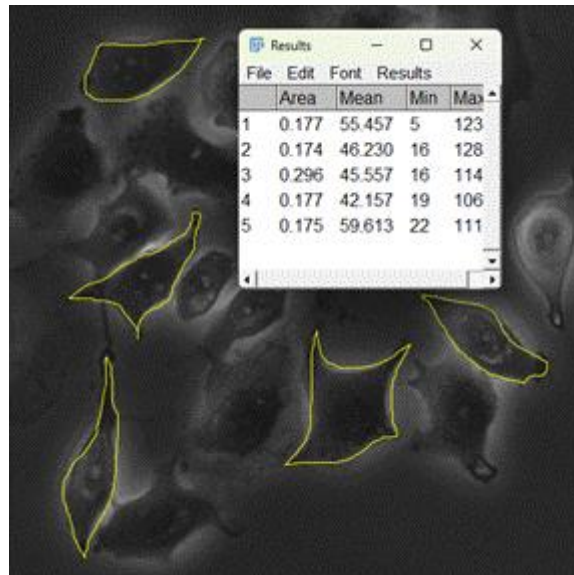
B5: 20 Resv



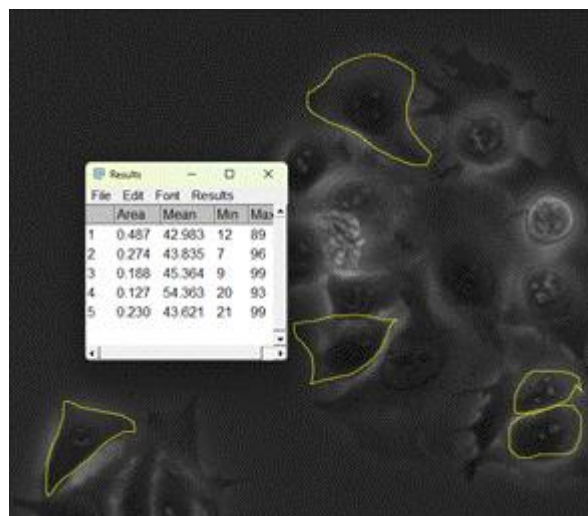
C5: 20 Resv +
NaBu



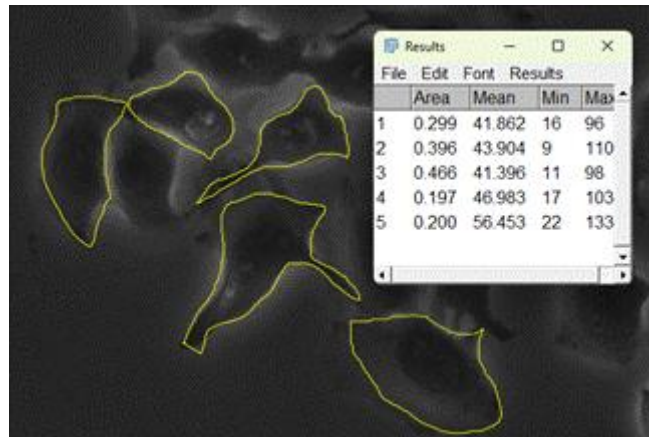
D5: 20 Resv +
NaBu



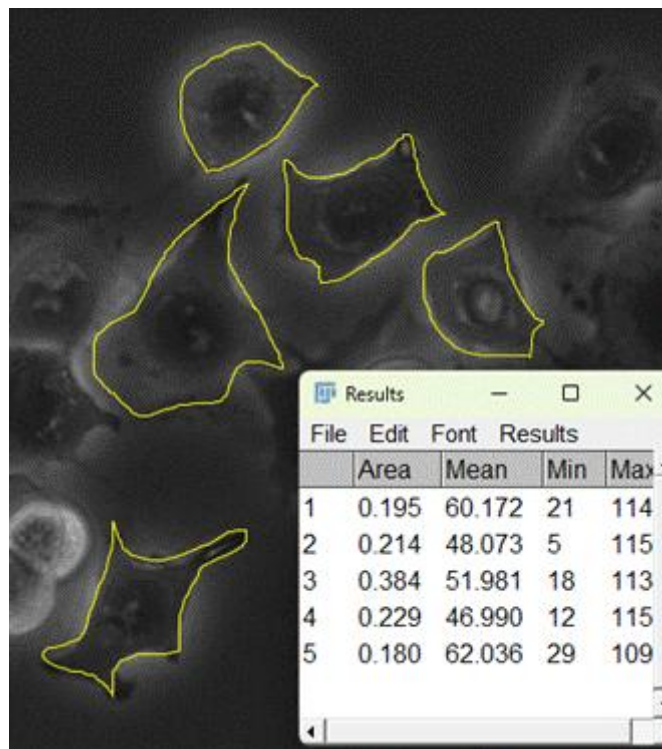
A6: 40 Resv



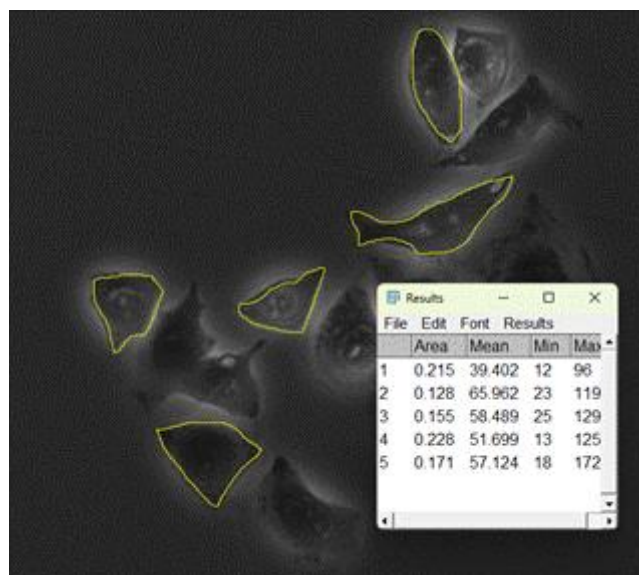
B6: 40 Resv



C6: 40 Resv + NaBu



D6: 40 Resv + NaBu



Determination of Growth rate

$$(t_2 - t_1) \cdot \left(\frac{\log(2)}{\log\left(\frac{Q_2}{Q_1}\right)} \right)$$

DLD-1:

t_1 = time when seeded (18 june)

t_2 =time when counted (22 june)

Q_1 = total number of cells seeded (18 june 500.000cells/ml was seeded)

Q_2 = total number of cells at counting. (22 june 1.131.180 cells/ml counted)*10ml
=11.311.800 cells

$$\frac{Q_2}{Q_1} = \frac{11311800}{500000} = 22,62$$

$$division\ time = 4 \cdot \left(\frac{\log(2)}{\log(22,62)} \right) = 0,888$$

$$0,888 \cdot 24 = 21,3$$

Caco-2:

t_1 = time when seeded (18 june)

t_2 =time when counted (22 june)

Q_1 = total number of cells seeded (18 june 500.000cells/ml was seeded)

Q_2 = total number of cells at counting. (22 june 195.075 cells/ml counted)*10ml =1.950.750
cells

$$\frac{Q_2}{Q_1} = \frac{1.950.750}{500000} = 3.90$$

$$division\ time = 4 \cdot \left(\frac{\log(2)}{\log(3.90)} \right) = 2.94$$

$$2.04 \cdot 24 = 48.9\ hours$$

Appendix: Lab journal

15 June

2020

Grön

Control

pro

Line

Long

Andreas

exp 1

exp 2

exp 3

exp 4

exp 5

exp 6

exp 7

exp 8

exp 9

exp 10

DLD-1

-Wash 2 times with 1
w/o Ca or Mg
-Trypsinate with 1ml trypsin-
EDTA
-Split cells
-Resuspend in 10 ml media
-Make new T75 for other days
0,5-1 million cells pr passage

Make mastermix for
experiment 1,2 and 3

500.000 cells +
(700.000*2)+500.000=
2.400.000 cells
20% extra is included)

0,5-1 million cells pr passage

Use $C1 \cdot V1 = C2 \cdot C2$ to
calculate/correct amounts of
cell-suspension for mastermix

Obs:

Colour: Orange-yellow

Confluence 75%.

Shape: not rounded

- We wash
DLD-1 medium with 10ml
bottle labeled PBS-1-
to wash the medium

- we tryptonise cells
and put in heat
for 7 min

- we were unsatisfied with results
and put back in heat for extra
2 min

- we split the cells mechanically 4 times
after being satisfied with tryptonise

- we checked if cells were single. We decided
to mechanically split a few more times. we checked
again, not satisfied and split few more times

notes: added a bit more

PBS medium than in
protocol

side 1

15/06/20

15-06-2016

- we start counting using hemocytometer.
using designated program given by lecturers

- we add our cell ^{DLD-1} suspension to salt water
concentration of 0.9%. 200 μ L. we do two
counts for accuracy.

cell count: 86 66

cell count: 87 07

- cells ^{suspension} used for counting is discarded

$$\frac{8666 + 8704}{2} = 8685 \text{ cells/0.5ml.}$$

$$8685 \cdot 51 \cdot 2 = 885.820 \text{ cells/ml}$$

New flask for further use on Thursday.
500.000 cells

$$\frac{500.000}{885.820} = 0.56 \text{ ml in new flask.}$$

- we start making the master mix

side 2

16-06-2025

DLD-7 ~ 20% confluence

Caco-2 about 20% confluence, attached to the bottom of the flask and looks healthy. Many single celled and beautiful

17/06-2026

DLD-7: Confluence is about not above 35%. attached to one another.

CACO2: also well grouped together. Around 20% - 30%.



18/06-2026: Confluence of Caco2 were around 40%. individual groups of around 10-20 cells stuck together.



Confluence of DLD-7 was around 50% with ~~groups~~ of individual groups of around 10-30 cells

side 3

18.06.2026

- BLD-1 is being manually split in order to prepare it for freezing.

- trypsin is added ~~and is~~ to the cells and incubated

side 4

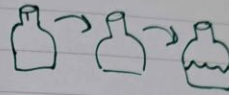
15 Juni 2026 green Gruppe

Wine
Lars
Andreas

Caco2	
- Wash 2 times with 10ml 85 mM sodium citrate - Trypsinate with 1ml trypsin-EDTA - Split cells - Resuspend in 10 ml media - Make new T75 for other days 0.2-0.5 million cells pr passage - Make mastermix for experiment 1,3 and 4 500.000 cells + 500.000 cells + 1.500.000 cells + 250.000 cell = 2.750.000 cells (20% extra is included) 0.2-0.5 million cells pr passage - Use $C1 \cdot V1 = C2 \cdot C210$ calculate correct amounts of cell-suspension for mastermix	

pre-trypsin
cell phenotype: red medium
~ 50% confluent
small clusters of cells

3x T75 flasks sucked & washed x2 w/ Na citrate



After trypsinisation we collected all cell suspension in 1x T75 flask and mechanically splitted them

before we transferred 200 µl in saline for cell count on the Coulter counter. We looked in the microscope to check the cells were single celled.

Counter 1: 11778

(0.5 ml sucked, 200 µl in 10 ml saline → x 102 factor)

Counter 2: 11995

avg: 11886.51.2

total 1.212.372 cells/ml

passage $300'000 \text{ cells} / (1.212'372 \text{ cells/ml}) = 0.247$

we take 250 µl → 303.093 cells

15/06/26

side

side 6

15 June 2026

exp 1:

Mastermix	Seed for experimental 1: NaBu dose response
20 ml	
500,000 cells	25,000 cell/mL, 1.0 mL/well in 24-well plate in column 1-4, 20 mL is enough for four columns. The rest of the plate is left empty.
	$C1 \cdot V1 = C2 \cdot V2$
	Leave ON in incubator

grau
more
line
Lars
Andreas

Case Study

DLD-11

MM: 20 mL - 0,56 = 19,44 mL

19,44 mL M60V

0,56 mL with DLD-1 cells

1200/76
side 11

16.06.2026 green team: Line, Lars, Andreas
DLD1
Experiment 1: NaBu dose response

Prepare dilutions of NaBu from 0.5-5 mM together with 0 mM (control with PBSw) in medium, at least 7 mL of each concentration. Keep the concentration of PBSw constant in each tube!

Cell Block	1	2	3	4	5	6
A						
B						
C						
D						

Sodium butyrate (NaBu); Stock: 1 M, dissolved in PBS

0.5 mM: 3.5 μ l 1M NaBu stock
1 mM: 7 μ l 1M NaBu stock
5 mM: 35 μ l NaBu stock

The largest volume is 35 μ l, so every tube should contain 35 μ l total PBS.

Prepare 7 mL of each condition

Final NaBu	1 M NaBu stock	PBSw added	Medium added	Final volume
0 mM (control)	0 μ l	35 μ l	6965 μ l	7000 μ l
0.5 mM	3.5 μ l	31.5 μ l	6965 μ l	7000 μ l
1 mM	7 μ l	28 μ l	6965 μ l	7000 μ l
5 mM	35 μ l	0 μ l	6965 μ l	7000 μ l

Incubate 48 hours.

- 8 tubes are taken and added 8 mL medium

- 2 are used: for control (40 μ l PBSw)
2 are used as 0.5 mM (4 μ l 1M NaBu + 36 μ l PBSw)
2 are used as 1 mM (8 μ l 1M NaBu + 32 μ l PBSw)
2 are used as 5 mM (40 μ l 1M NaBu)

1 M NaBu	PBSw	Final vol
0	40 μ l	8000 μ l
4 μ l	36 μ l	-11-
8 μ l	32 μ l	-11-
40 μ l	0	-11-

~~The stock is seeded~~

- Treated cells with NaBu in different concentrations

0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

Control 0.5 1 5

- Old medium is replaced with NaBu medium

16/06/2026

Side ce

DLD-1

17.06.2026 green team: Line, Lars, Andreas
DLD1
Experiment 1: ~~NaBu dose-response~~

Cell Freezing:

- Check cells for good growth (Non confluent)
- Remove old media using pasteur pipettes
- Add PBSw/o to carefully wash, removed with suction x2
- Trypsinize cells
- Resuspend cell in PBSmedia (not the same way, TBA). Centrifuge at 145 G for 3-5 min at room temp.
- Remove media with Pasteur pipettes
- Resuspend in freezing media (McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol) 1mL per cryotube (Often cells are frozen at very high densities: $> 2 \times 10^6$ per cryo-vial. This ensures speedy recovery of cells following thawing.)
- Transfere 1 mL per cryotube
- Placed in "mr freeze" container and placed at -80°C . After at least 3h, place it in cryo containing liquid N₂.
- Check success by thawing a frozen culture

cells was not frozen today (17/06/2026) because the cells wasnt confluent enough.

→ DMSO

side 13 14
17/06/2026

DLD-1

18.06.2026 green team: Line, Lars, Andreas
DLD1
Experiment 1: NaBu dose response

1.1. Exp 1 freeze and passage (if we don't freeze caco)

Cell Freezing:

- Check cells for good growth (Non confluent)
- Remove old media using pasteur pipettes
- Add PBS w/o to carefully wash, removed with suction x2
- Trypsinize cells
- Resuspend cell in ~~DMEM~~ media (not the same way, TBA). Centrifuge at 145 G for 3-5 min at room temp.
- Remove media with Pasteur pipettes
- Resuspend in freezing media (McCoy 5A + 10% FBS + 1% P/S + 10% Glycerol) 1 mL per cryotube (Often cells are frozen at very high densities: $> 2 \cdot 10^6$ per cryo-vial. This ensures speedy recovery of cells following thawing.)
- Transfer 1 mL per cryotube
- Placed in "mr freeze" container and placed at -80°C . After at least 3h, place it in cryo containing liquid N2.
- Check success by thawing a frozen culture

1.2 Image and count cells Perhaps passage

- Acquire phase contrast pictures kl 9.45.
- Remove only one 24-well plate at a time from the incubator
- Take pictures in the center of each well to obtain optimal phase contrast. Use 20x magnification and adjust phase ring to 10x/20x.
- Wash cells 2x with 1 mL PBS w/o or SC. add 0,2 mL trypsin.
- Incubate for 20min
- add 0.8 mL PBS w/o. Resuspend on ice
- Transfer 0.8 mL cell suspension to 10 mL counting solution (0.9% saline)
- Estimate cell number using Coulter Counter. Count (above $9 \mu\text{m}$)
- Calculate cells/mL (dilution factor is $800 \mu\text{L} / 10800 \mu\text{L}$) (Total volume used for counting is 0.5 mL)

- Determine Cell area (Acquire picture of each well, cover at least 10 cells, 20x magnification)

cell obs: orange medium
~ 50% confluence

counts: $4884 + 4829$ avg $\rightarrow 4852$
total $200 \mu\text{L} \rightarrow \times 100 \text{ dil}$
 497.913 cells/mL

- 300g

resuspend in new media for counting
- DMSO

$2 \cdot 10^6 \text{ cells} / 497.913 \text{ cells/mL}$

8.75

~~8.75~~ mL spun $\rightarrow 4756738 \text{ cel}$

want $2 \cdot 10^6 \text{ cells/mL}$ cryosolution

use e.g. 2.2 mL $\rightarrow 4.4 \cdot 10^6 \text{ cells}$

make 2.1 mL resuspension instead

resusped centrifuged cells until 0.2 mL

left & resuspended in 2.1 mL freezing

media (freezing media: 2.7 mL McCoy &

0.3 mL DMSO)

each cryotube received 1 mL

resuspension

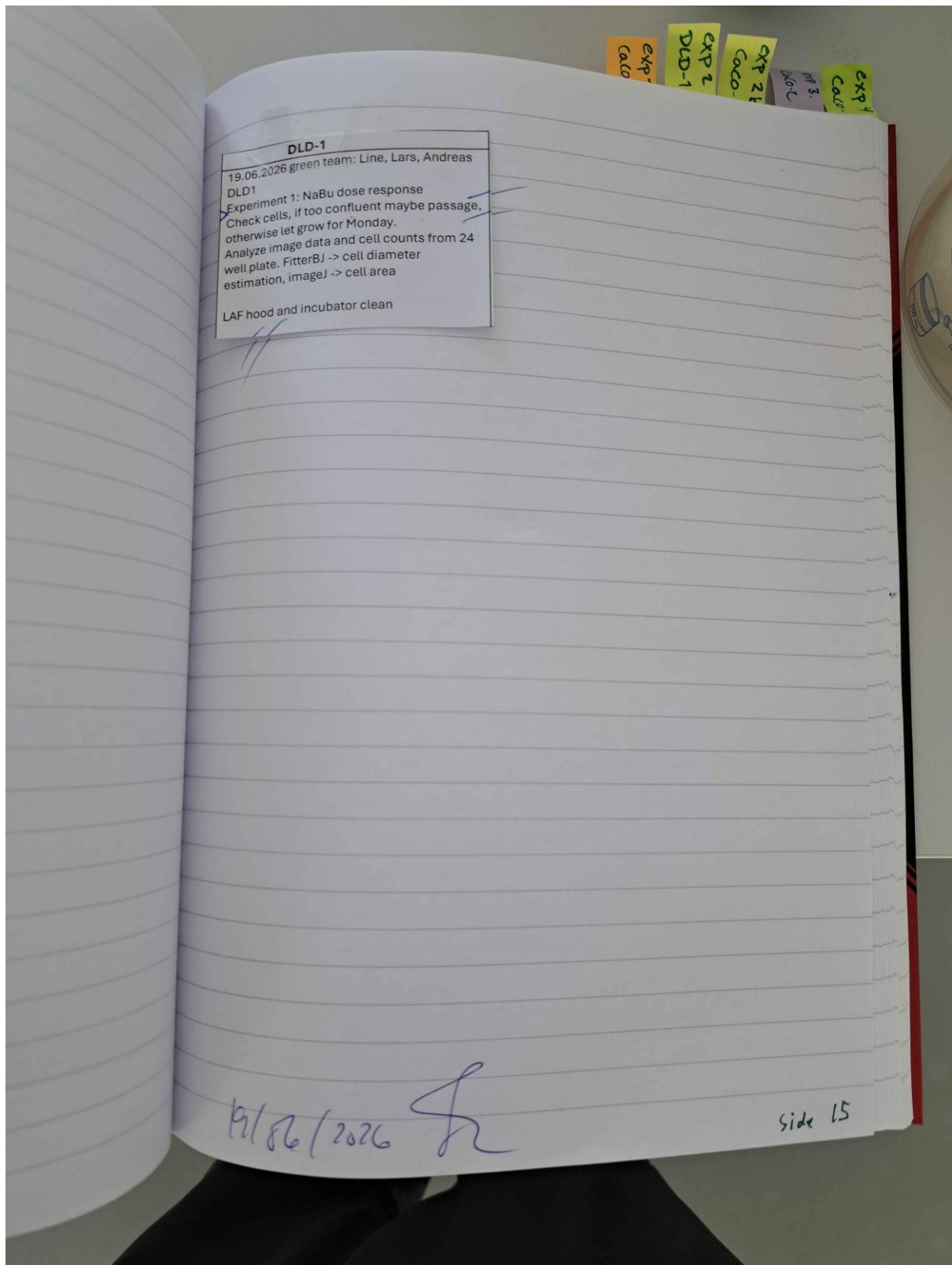
12.5 fold

$73.5 \times \text{dil}$

R

2/6/26

side 14



DLD-1

22.06.2026 green team: Line, Lars, Andreas
DLD1
Experiment 1: NaBu & Resv dose response
DLD only

Use NaBu concentration of 1mM
Seed 2 x 24 well-plate with 25.000 cell/ml,
1 ml/well.
Pre-wet wells with 0.5 ml McCoy before
adding 1 ml 25'000 cell/ml
Mastermix of 55 ml. (1'375'000 cells
McCoy medium)

Mastermix's of DLD:

Exp1: 55 ml with 1.375.000 cells
Exp 2a: 36 ml 2.100.000 cells ✓
Exp 2b: 10 ml with 500.000 cells
Exp 3: 30 ml with 1'500'000 cells
Exp 4: -

New passage
500.000

$$\frac{1.131.180}{500.000} = 0,442 \text{ ml cell sep for new passage}$$

560.000 → New passage
5.975.000 cells in total

11531 and 10649 pr 0,5 ml in coulter counter

$$\frac{11531 + 10649}{2} = 11090 \text{ Average} \cdot 51 \cdot 2 = 1.131.180 \text{ cells/ml.}$$

$$\frac{1.375.000 \text{ cells}}{1.131.180 \text{ cells/ml}} = 1,215 \text{ mL} \cdot 2 = 2,43 \text{ mL} \cdot 0,60777 \text{ mL}$$

$$\text{Exp} \quad 55 \text{ mL} - 1,2 = 53,78 \text{ mL} \quad 55 \text{ mL} \cdot 0,2 = 27,5 \text{ mL}$$

$$\text{Exp 3: } \frac{1.500.000 \text{ cells}}{1.131.180 \text{ cells/mL}} = 1,326 \text{ mL}$$

Obs: i experiment 1 laver vi i stedet 2 tubes
med 27,5 mL i hver fordi vores tubes kan
ilde rumme 55 mL. (26,9 mL medie + 0,6 cell^{sup})
x2

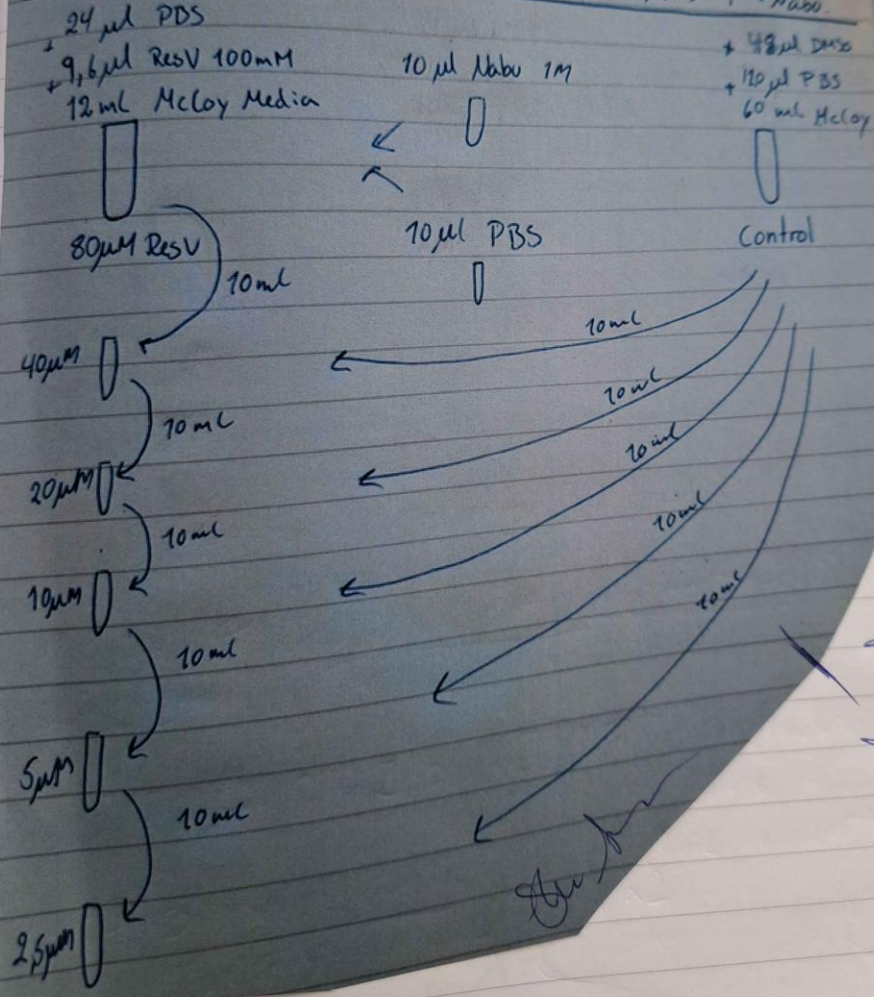
27/06/2026

60

27-06-2026

Green team, Lars, Line, Andreas
 experiment 1 week 2 Nabu + Resv treatment
 (DLD-7)

1						
A	control	2,5µM Resv	5µM Resv	10µM Resv	20µM Resv	40µM Resv
B	-11-	-11-	-11-	-11-	-11-	-11-
C	Nabu	2,5µM Resv	5µM Resv	10µM Resv	20µM Resv	40µM Resv
D	Nabu	+ Nabu	+ Nabu	+ Nabu	+ Nabu	+ Nabu



DLD-1
 24.06.2026 green team: Line, Lars, Andreas
 DLD1
 Experiment 1: NaBu dose response

Input: Two 24-well plate.

	1	2	3	4	5	6
A	Control	2.5 μ M Resv	5 μ M Resv	10 μ M Resv	20 μ M Resv	40 μ M Resv
B						
C	NaBu	2.5 μ M Resv + NaBu	5 μ M Resv + NaBu	10 μ M Resv + NaBu	20 μ M Resv + NaBu	40 μ M Resv + NaBu
D						

Workflow:

- Do nothing for today. Maybe check cells. It is incubating for Thursday.

Input: Frozen cells

Thawing Workflow:

- Preheat 15 ml pre-heated media (McCoy) in T75 flask.
- Extract cryotube from cry-container. Do not make the tube explode. Transport to lab, e.g. in lab coat pocket.
- Clean cryotube in ethanol. (Can you handle this with regular gloves?)
- Transfer pre-heated media to tube. Resuspend. Repeat until the cells are out.
- Put them incubator.

= also
 check &
 possibly
 passage
 cells

confluent: 40%, media light orange

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25-06-2026

DLD-1

25.06.2026 green team: Line, Lars, Andreas
DLD1

Experiment 1: NaBu dose response

1.2 Image and count cells

- Acquire phase contrast pictures kl 8.30.
- Remove only one 24-well plate at a time from the incubator
- Take pictures in the center of each well to obtain optimal phase contrast. Use 20x magnification and adjust phase ring to 10x/20x.
- Wash cells 2x with 1 mL PBS w/o or SG. add 0,2 mL trypsin.
- Incubate for 20min
- add 0.8 mL PBS w/o. Resuspend on ice
- Transfer 0.8 mL cell suspension to 10 mL counting solution (0.9% saline)
- Estimate cell number using Coulter Counter. Count (above 9 μ m)
- Calculate cells/mL (dilution factor is 800 μ L / 10000 μ L) (Total volume used for counting is 0.5 mL) - 13.5 x dilution factor
- Determine Cell area (Acquire picture of each well, cover at least 10 cells, 20x magnification)

main T75 culture:

media orange
confluence 70%

thawed culture:

dead cells floating in media
live cells adhering to flasks
 $\approx$ 15% confluence

0.2 mL
0.8 mL

200 01/70200 01

OBS. Sample A1 on the
second well plate got

2 times trypsin and would therefore be more
diluted. 0,8 mL trypsin

- 1 0,2 mL trypsin
- 0,8 mL trypsin
- * 0,8 mL PBS.

passaging: thought i got all cells during trypsin
later saw many cells left, so didn't get many out
count 2049 $\sim$ 208.998 cells/mL
seed 500'000 cells / 208.998 cells/mL
= 2.4 mL

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Caco2
 22.06.2026 green team: Line, Lars, Andreas
 CaCo-2
 Passage 300'000 cells
 Experiment 1: NaBu dose response
 No Caco for exp 1 week 2 dose response

Mastermix's of Caco2:
 Exp1: -
 Exp 2a: -
 Exp 2b: 10 ml with 500.000 cells
 Exp 3: -
 Exp 4: -

obs: Confluence
 check: ~~100%~~ 95%
 - it looks like
 skin surface
 - we borrow another
 culture due to
 the high confluence.
 the new culture has
 a confluence of ~~90-95%~~
 35%.

Passage 22 caco2

Coulter counter Caco2

$$\begin{array}{l} 9: 1614 \\ 2: 2211 \end{array} \rightarrow \frac{1614}{2211} \cdot \frac{1614 + 2211}{2} = 1912.5$$

$$1912.5 \cdot 51.2 = 195.075 \text{ cells/mL}$$

$$\frac{300\,000 \text{ cells}}{195.075 \text{ cells/mL}} \sim 1.537 \text{ mL}$$

$$10 \text{ mL} - 1.537 \text{ mL} =$$

12/06/2026 *R*

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Caco2

25.06.2026 green team: Line, Lars, Andreas
CaCo-2

Experiment 1: NaBu dose response

Image and count cells

- Acquire phase contrast pictures kl 8.30.
- Remove only one 24-well plate at a time from the incubator
- Take pictures in the center of each well to obtain optimal phase contrast. Use 20x magnification and adjust phase ring to 10x/20x.
- Wash cells 2x with 1 mL PBS w/o or SC. add 0,2 mL trypsin.
- Incubate for 20min
- add 0.8 mL PBS w/o. Resuspend on ice
- Transfer 0.8 mL cell suspension to 10 mL counting solution (0.9% saline)
- Estimate cell number using Coulter Counter. Count (above 9 μ m)
- Calculate cells/mL (dilution factor is 800 μ L/10800 μ L) (Total volume used for counting is 0.5 mL) - 13.5 x dilution factor
- Determine Cell area (Acquire picture of each well, cover at least 10 cells, 20x magnification)

main TDS culture
confluent 70+80%
maybe too late to split

Caco2 refuse to split well
we carry on with some
clumps

count 2080×102

= 212.160 cells/mL

seed 300'000 $\rightarrow$

$300'000 / 212160 = 1.414$ mL

seed 1,4 mL

fill with 9 mL DMEM